

IIT Jodhpur IMAT BS - Question Bank

Time: 2 Hours | Maximum Marks: 198

Test Pattern

- **VARC:** 24 questions | 40 min
- **DILR:** 18 questions | 40 min
- **QA:** 24 questions | 40 min
- **Total:** 66 questions across 3 sections | Duration 2 Hours | Maximum Marks 198

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Each section is allotted 40 minutes; the total duration is 2 Hours.
- Attempt the sections in the given order and manage your time across all sections.

This question bank holds 198 questions -- BS 198 (72 VARC/54 DILR/72 QA). Each test is assembled from it following the pattern above. Correct option shown in green.

Section: VARC (40 min)

Instructions (BS1-VARC-RC1): study the shared passage below and answer its 4 question(s).

For most of human history, the night sky was a shared inheritance: a band of stars visible to anyone who looked upward. That inheritance is quietly dissolving. The spread of electrified cities has scattered artificial light into the atmosphere, where dust and moisture scatter it further, producing a diffuse glow that astronomers call skyglow. Under this veil, the faint light of distant stars is simply drowned out. Surveys now suggest that a large share of people alive today have never seen the Milky Way from where they live, and that the brightness of the night sky over inhabited regions is rising by a few per cent each year.

The consequences reach well beyond stargazing. Many organisms time their behaviour by natural cycles of darkness. Migrating birds, which navigate partly by the stars, can be disoriented by illuminated buildings; insects exhaust themselves circling lamps; hatchling sea turtles, which instinctively crawl toward the brightest horizon, may turn inland toward roads instead of out to sea. In each case a cue that was reliable for millions of years has been overwritten by a human signal.

What makes light pollution unusual among environmental problems is how readily it can be reversed. Carbon dioxide lingers in the atmosphere for centuries; a poorly aimed streetlamp stops polluting the moment it is switched off or shielded. Lighting that points downward, that is no brighter than a task requires, and that is dimmed when no one is present can preserve both safety and darkness. The obstacle is not technical but a habit of mind that equates more light with more progress. Recovering the night sky, then, asks less for new invention than for a willingness to use, more thoughtfully, the light we already command.

Q1. [BS1-VARC-RC1 · mcsc · diff 0.5]

Which of the following best expresses the main idea of the passage?

- A) Artificial light has now made it completely impossible for most people on Earth to see any stars at all from any of the places where they currently live and work.
- B) The disorientation of migrating birds, circling insects, and stranded sea turtles is now the single most urgent environmental crisis facing the entire planet.
- C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology. ✓
- D) Astronomers, rather than elected officials, should be given final authority over urban lighting decisions because they understand skyglow better than city planners.

Answer: C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology.

Q2. [BS1-VARC-RC1 · mcsc · diff 0.5]

According to the passage, the author makes all of the following claims EXCEPT:

- A) Skyglow is produced partly because dust and moisture in the atmosphere scatter artificial light.
- B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented. ✓
- C) Some animals rely on natural darkness or starlight as a behavioural cue that artificial light can disrupt.
- D) The brightness of the night sky over inhabited regions has been increasing year by year.

Answer: B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented.

Q3. [BS1-VARC-RC1 · mcsc · diff 0.5]

Why does the author contrast light pollution with carbon dioxide?

- A) To argue that carbon dioxide is, in the end, a far less serious threat to migrating wildlife than artificial light is.
- B) To suggest that the two problems are caused by exactly the same kind of polluting industrial activity in modern cities.
- C) To prove that most environmental problems generally cannot be solved at all without first achieving major scientific breakthroughs.
- D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time. ✓

Answer: D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time.

Q4. [BS1-VARC-RC1 · mcsc · diff 0.5]

It can be inferred that the author would most likely agree with which of the following statements?

- A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around. ✓
- B) The loss of the visible Milky Way is, regrettably, an entirely unavoidable cost of building any modern, fully electrified society.
- C) Cities ought to abandon their street lighting almost entirely in order to recover the lost visibility of the dark and starry night sky.
- D) Concern about the fading night sky is mainly a private preoccupation of professional astronomers rather than a truly broad public issue.

Answer: A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around.

Instructions (BS1-VARC-RC2): study the shared passage below and answer its 4 question(s).

When a result in science is described as 'replicated,' it means that an independent team, following the same procedure, has obtained a similar finding. Replication is often called the gatekeeper of reliable knowledge, because a single study, however carefully done, can be misled by chance, by an unusual sample, or by an undetected flaw in its design. Only when a result reappears under different hands does the community gain confidence that it reflects something real rather than an accident of one laboratory.

In recent years, however, attempts to repeat well-known studies have frequently failed to reproduce the original effects. Several factors contribute. Researchers face strong incentives to publish striking, novel results, while careful repetitions of someone else's work attract little reward and are seldom published. Small samples can yield impressive-looking outcomes that vanish when the study is enlarged. And the freedom to analyse data in many ways allows a researcher, even unintentionally, to keep adjusting the analysis until a pattern emerges.

None of this means that science is broken. The very fact that the failures are being documented is itself a product of the scientific method turning its scrutiny on its own practices. Reforms are already spreading: some journals now invite researchers to register their hypotheses and planned analyses before collecting data, and commit to publishing the results whether or not they are striking. Others reserve space specifically for replication studies. What looks at first like a crisis of confidence may be better understood as a discipline learning, in public, to hold itself to its own standards.

Q5. [BS1-VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Scientific journals are the primary cause of unreliable published findings and therefore should no longer be trusted at all by readers.
 - B) Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method. ✓
 - C) Small samples are, on their own, the single underlying reason that so many famous studies have proven difficult to reproduce reliably.
 - D) Replication is essentially impossible in actual practice because no two separate laboratories ever follow exactly the same detailed procedure.
- Answer: B)** Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method.

Q6. [BS1-VARC-RC2 · mcsc · diff 0.5]

The author identifies all of the following as factors contributing to replication failures EXCEPT:

- A) The freedom to analyse data in many ways until a pattern happens to appear.
 - B) A deliberate effort by most researchers to fabricate data in order to deceive their peers. ✓
 - C) Incentives that reward novel findings while offering little reward for repeating others' work.
 - D) Small samples that can produce striking results which disappear when the study is enlarged.
- Answer: B)** A deliberate effort by most researchers to fabricate data in order to deceive their peers.

Q7. [BS1-VARC-RC2 · mcsc · diff 0.5]

What is the main purpose of registering hypotheses and planned analyses before collecting data, as described in the passage?

- A) To guarantee that each and every registered study will reliably end up producing some kind of novel and publishable finding.
 - B) To replace careful replication studies entirely, so that any repetition of other people's earlier work is no longer needed at all.
 - C) To allow researchers to choose freely, after they have seen the data, which analysis presents their results most favourably.
 - D) To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way. ✓
- Answer: D)** To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way.

Q8. [BS1-VARC-RC2 · mcsc · diff 0.5]

It can be inferred from the passage that the author regards the documentation of replication failures as:

- A) definite proof that most published scientific results will, sooner or later, be shown to be entirely false.
 - B) an unfortunate and costly distraction that needlessly diverts attention away from producing genuinely novel discoveries.
 - C) a serious problem that can only ever be solved by abandoning the basic requirement of replication altogether.
 - D) evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay. ✓
- Answer: D)** evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay.

Instructions (BS1-VARC-RC3): study the shared passage below and answer its 4 question(s).

A city map is never a perfect copy of the city it represents. If it were, it would have to be exactly as large and as complicated as the city itself, and so would be useless to a traveller. The value of a map lies precisely in what it leaves out. By dropping countless details and keeping only roads, rivers, and a few landmarks, a map becomes small enough to hold in the hand and clear enough to guide a journey.

The same principle applies to scientific models. A model of the weather, or of a population of animals, does not try to capture every raindrop or every individual creature. Instead it keeps the features that matter for the question being asked and ignores the rest. A model built to predict tomorrow's temperature can safely leave out the colour of people's umbrellas; a model of how a disease spreads can ignore the brand of car that infected people drive. Choosing what to leave out is not a weakness of a model but the very thing that makes it useful.

This has a practical lesson. When we judge a model, the right question is not 'Is it complete?' but 'Is it useful for its purpose?' A simple model that answers the question at hand may be far more valuable than an elaborate one that tries to include everything and so becomes impossible to understand. As one saying among scientists puts it, all models are wrong, but some are useful. Wisdom lies in matching the simplicity of the model to the job it must do.

Q9. [BS1-VARC-RC3 · mcsc · diff 0]

Which of the following best captures the main idea of the passage?

- A) Scientific models should always try to include as much fine-grained detail as they possibly can in order to be truly accurate.
- B) Maps are, in the end, far more reliable guides to a city than scientific models can ever be to the natural world.
- C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose. ✓
- D) The colour of people's umbrellas and the brand of a car are in fact important variables in both weather and disease models.

Answer: C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose.

Q10. [BS1-VARC-RC3 · mcsc · diff 0]

Why does the author say that a perfectly complete map would be useless?

- A) Because maps are always rather less accurate than the kind of scientific models that are discussed later on in the passage.
- B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller. ✓
- C) Because real cities change far too quickly for any printed map of them to remain fully correct for very long.
- D) Because no ordinary traveller is ever genuinely interested in the roads, the rivers, or the landmarks of a city.

Answer: B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller.

Q11. [BS1-VARC-RC3 · mcsc · diff 0.5]

It can be inferred that the author would most likely judge a scientific model by asking:

- A) whether it serves its intended purpose well, even if it leaves out many features of the real situation. ✓
- B) whether it manages to include every single variable that could conceivably affect the final outcome in any way.
- C) whether it is noticeably more detailed and elaborate than any competing model that addresses the same question.
- D) whether it perfectly reproduces the whole of reality without leaving out or omitting anything at all whatsoever.

Answer: A) whether it serves its intended purpose well, even if it leaves out many features of the real situation.

Q12. [BS1-VARC-RC3 · mcsc · diff 0.5]

What is the main function of the saying 'all models are wrong, but some are useful' in the passage?

- A) To warn readers firmly that scientific models really should not be trusted when making any practical decision.
- B) To introduce a fresh objection that the author then goes on to refute carefully and in some detail.
- C) To claim that maps, quite unlike scientific models, are essentially never wrong about the cities they represent.
- D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose. ✓

Answer: D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose.

Q13. [BS1-VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Coral reefs occupy less than one percent of the ocean floor, yet they shelter roughly a quarter of all marine species.
- (2) The dense limestone structures act as nurseries where countless fish spawn and mature.
- (3) They also buffer nearby coastlines by absorbing much of the energy carried by incoming waves.
- (4) Many tropical resorts now offer guided diving tours that bring tourists close to these formations.
- (5) Because of this ecological richness, the loss of a single reef can ripple through an entire marine food web.

Which sentence is the odd one out?

- A) 3
- B) 5
- C) 4 ✓
- D) 1
- E) 2

Answer: C) 4

Q14. [BS1-VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) The printing press, introduced in fifteenth-century Europe, dramatically lowered the cost of reproducing texts.
- (2) Books that had once taken months to copy by hand could now be produced in days.
- (3) As prices fell, reading spread beyond monasteries and universities to ordinary townspeople.
- (4) This wider circulation of ideas helped fuel movements in science and religion.
- (5) Modern libraries increasingly lend electronic books that readers download to personal devices.

Which sentence is the odd one out?

- A) 5 ✓
- B) 1
- C) 2
- D) 3
- E) 4

Answer: A) 5

Q15. [BS1-VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Honeybees communicate the location of food through a movement known as the waggle dance.
- (2) Beekeepers harvest surplus honey by carefully removing frames from the hive.
- (3) A returning forager runs in a figure-eight pattern on the vertical comb.
- (4) The angle of the run relative to gravity encodes the direction of the food source.
- (5) The duration of the waggle phase signals how far away the food lies.

Which sentence is the odd one out?

- A) 5
- B) 2 ✓
- C) 1
- D) 4
- E) 3

Answer: B) 2

Q16. [BS1-VARC-JUMBLE-1 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. The result was a network of stone roads stretching across the whole empire.
2. Ancient Rome needed a fast way to move its armies between distant provinces.
3. To meet this need, engineers laid down durable paved routes wherever the legions marched.
4. Centuries later, many of these same roads still guide the paths of modern highways.

Select the correct sequence of sentences.

- A) 2341
- B) 3214
- C) 2314 ✓
- D) 2134

Answer: C) 2314

Q17. [BS1-VARC-JUMBLE-2 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. Over time, this trapped heat raises the planet's average surface temperature.
2. Certain gases in the atmosphere allow sunlight to pass through but absorb the heat that the ground radiates back.
3. This natural process is what scientists call the greenhouse effect.
4. When the concentration of these gases rises, more heat is retained than before.

Select the correct sequence of sentences.

- A) 2341
- B) 3241
- C) 2314 ✓
- D) 2431

Answer: C) 2314

Q18. [BS1-VARC-JUMBLE-3 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. She began by sketching rough thumbnails to test different compositions.
2. Maria had been asked to design the cover for the school magazine.
3. Once a layout felt right, she refined it into a detailed digital drawing.
4. Finally, she added colour and sent the finished file to the printer.

Select the correct sequence of sentences.

- A) 1234
- B) 2143
- C) 2134 ✓
- D) 2314

Answer: C) 2134

Q19. [BS1-VARC-JUMBLE-4 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. At first the seedlings were too fragile to survive outdoors.
2. The gardeners started the tomato plants indoors during early spring.
3. So they kept the young plants under warm lamps for several weeks.
4. Only when the stems had thickened did the gardeners move them into the open beds.

Select the correct sequence of sentences.

- A) 2314
- B) 1234
- C) 2134 ✓
- D) 2143

Answer: C) 2134

Q20. [BS1-VARC-SUMMARY-1 · mcsc · diff 0]

Public libraries today do far more than lend books. Many now offer free internet access, run job-search workshops, host story hours for children, and lend tools or musical instruments. For people without these resources at home, the library has become a community centre as much as a place to read.

Which of the following best summarises the paragraph?

- A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books. ✓
- B) Public libraries today mainly function as places that lend out musical instruments and household tools to the local residents who ask.
- C) Libraries have lost all interest in books and now devote themselves entirely to providing free internet access to their local visitors.
- D) The single most valuable thing libraries do now is to host the enjoyable weekly story hours for the young children of the community.

Answer: A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books.

Q21. [BS1-VARC-SUMMARY-2 · mcsc · diff 0.5]

Sleep researchers have found that pulling an all-nighter before an exam usually backfires. While extra waking hours seem to allow more study, the tired brain struggles to store new information, and fatigue slows recall the next day. Students who sleep a full night generally remember more, even though they spent fewer hours reviewing.

Which of the following best summarises the paragraph?

- A) Students who choose to stay awake all night before an exam manage to study far more material and so tend to recall it better.
- B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better. ✓
- C) How much a student sleeps has no measurable effect at all on how well that student later performs on a difficult written examination.
- D) Sleep researchers now firmly advise students that they should never review any of their own notes during the evening before an exam.

Answer: B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better.

Q22. [BS1-VARC-SUMMARY-3 · mcsc · diff 0.5]

Electric cars produce no exhaust as they drive, which clearly improves air quality in cities. Yet whether they are truly cleaner depends on how their electricity is generated. Where power comes mostly from coal, the overall emissions can rival those of petrol cars; where it comes from renewables, the advantage is large.

Which of the following best summarises the paragraph?

- A) Coal-fired power stations still remain the main source of almost all of the electricity that is used to charge most electric cars today.
- B) Electric cars are always far cleaner than ordinary petrol cars in every situation, no matter how their electrical power happens to be generated.
- C) Electric cars release no pollution at all from their tailpipes while they are being driven along busy city streets and the open roads beyond.
- D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced. ✓

Answer: D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced.

Q23. [BS1-VARC-SUMMARY-4 · mcsc · diff 0.5]

Volunteering is often described as pure generosity, but studies suggest the giver gains too. People who regularly help others report lower stress, a stronger sense of purpose, and wider social networks. In this sense, the act of giving time quietly returns a benefit to the volunteer.

Which of the following best summarises the paragraph?

- A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits. ✓
- B) Volunteering is best understood as a wholly selfless act of giving that quietly returns no benefit of any real kind to the person doing it.
- C) People generally choose to volunteer mainly because doing so lets them expand and strengthen their own valuable personal social networks.
- D) Regular volunteers consistently report that they now feel considerably less stressed and rather more relaxed than they did before they began.

Answer: A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits.

Q24. [BS1-VARC-SUMMARY-5 · mcsc · diff 0.5]

When a new word enters a language, it rarely arrives by official decree. Instead, speakers begin using it, others copy them, and dictionaries record the word only after it has already taken hold. Language, in other words, is shaped from the bottom up by ordinary use rather than handed down by authorities.

Which of the following best summarises the paragraph?

- A) Dictionaries hold the real authority here and are the bodies responsible for officially deciding which new words a language may properly adopt.
- B) Most new words manage to enter a language only after some official government body has formally reviewed and then finally approved each one.
- C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time. ✓
- D) Speakers occasionally invent striking new words that many other people across the wider language community then choose to copy and reuse.

Answer: C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time.

Instructions (BS2-VARC-RC1): study the shared passage below and answer its 4 question(s).

****Read the following passage carefully and answer the question that follows.****

Before the middle of the nineteenth century, most towns kept their own local time, setting clocks by the moment the sun stood highest overhead. A traveller moving west by carriage might find each new town a few minutes behind the last, and nobody minded, because journeys were slow enough that small differences never mattered. Railways changed that arithmetic. A train that left one city at noon and reached another whose clocks also read noon had, on paper, taken no time at all. Timetables became puzzles, and collisions on single-track lines were a real danger when two crews disagreed about when a siding had to be cleared.

The response was not a government decree but a private one. Railway companies began publishing schedules based on a single reference clock, usually the time kept at their largest terminus, and asked stations along the line to follow it. Towns that resisted found themselves consulting two clocks: one on the church tower for daily life and one at the station for departures. In practice the station clock won, because it governed the arrival of mail, freight and visitors.

Only later did legislatures ratify what commerce had already settled, dividing territories into broad zones within which every clock agreed. The reform is often described as a triumph of science, and astronomers did supply the measurements, but the pressure came from the ledger rather than the observatory. What made local time intolerable was not that it was inaccurate; by the standard of the sun it was perfectly accurate. It was that accuracy of that kind could not be shared. A network needs its members to agree on a convention, and the value of the convention lies almost entirely in the agreement, not in the truth of the thing agreed. That is why the change, once begun, spread quickly: each town that joined made joining more attractive for the rest.

Q25. [BS2-VARC-RC1 · mcsc · diff 0.5]

Which of the following best captures the central point of the passage?

- A) Uniform clock time spread because travellers grew tired of resetting their watches in every town they entered.
- B) Uniform clock time spread because networks needed a shared convention, not because local time was wrong. ✓
- C) Uniform clock time spread because governments saw that public safety required identical clocks everywhere.
- D) Uniform clock time spread because astronomers at last produced measurements accurate enough to trust.

Answer: B) Uniform clock time spread because networks needed a shared convention, not because local time was wrong.

Q26. [BS2-VARC-RC1 · mcsc · diff 0.5]

The passage attributes all of the following to the arrival of railways EXCEPT:

- A) confusion in drawing up timetables that different lines could actually work with
- B) danger of accidents when two crews disagreed about when a track would be clear
- C) a legal rule obliging church towers to show the same hour as the station clock ✓
- D) schedules published from a single reference clock kept at a large terminus

Answer: C) a legal rule obliging church towers to show the same hour as the station clock

Q27. [BS2-VARC-RC1 · mcsc · diff 0.5]

It can be inferred from the passage that a town's church clock lost its authority mainly because:

- A) astronomers had declared solar time an unscientific way of setting clocks
- B) residents faced penalties for keeping time that differed from the railway's
- C) the church clock was found to be measurably slower than the railway's own clock
- D) the station clock decided when mail, goods and visitors would actually arrive ✓

Answer: D) the station clock decided when mail, goods and visitors would actually arrive

Q28. [BS2-VARC-RC1 · mcsc · diff 0.5]

The author calls local time 'perfectly accurate' in order to:

- A) suggest that the towns were right to resist the schedules imposed on them
- B) concede that early railway schedules were built on faulty measurements
- C) praise the astronomers who first measured the sun's position from each town
- D) stress that its defect was not error but that it could not be shared ✓

Answer: D) stress that its defect was not error but that it could not be shared

Instructions (BS2-VARC-RC2): study the shared passage below and answer its 4 question(s).

****Read the following passage carefully and answer the question that follows.****

Beneath the floor of most forests lies a dense mesh of fungal threads, thinner than hair and far more numerous than roots. Many of these fungi wrap themselves around, or grow into, the roots of trees. The arrangement suits both sides: the fungus takes sugars that the tree makes in its leaves, and the tree gains water and minerals that the fungal threads, with their enormous surface area, can gather from soil far beyond a root's reach.

Because a single fungus can be linked to several trees at once, the threads also form a route between plants. Experiments in which one tree is fed carbon labelled so that it can be traced show that a small share of that carbon later turns up in neighbours connected to the same fungal partner. This finding has been retold in popular writing as evidence that forests behave like cooperatives, with vigorous trees deliberately feeding shaded seedlings.

The evidence supports something narrower. That a substance moves between two organisms does not establish that either one intended the transfer, or that the recipient benefited enough to matter. Carbon may simply follow a gradient, flowing from where it is abundant to where it is scarce, much as water does. The fungus, moreover, is not a neutral pipe. It is an organism with interests of its own, and it may retain much of what passes through, trading resources on terms that favour itself rather than any particular tree.

None of this makes the network unimportant. A seedling that gains even a modest supply of minerals in its first seasons may survive a drought it would otherwise not survive, and that is a large effect from a small transfer. The caution concerns the story we drape over the measurement. Describing the forest as generous is a metaphor; the data describe an exchange whose accounting has not yet been done.

Q29. [BS2-VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the main argument of the passage?

- A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data. ✓
- B) Fungal links between trees mostly injure the trees, because the fungus keeps nearly every resource.
- C) Fungal links between trees are so faint that the transfers recorded are probably measurement error.
- D) Fungal links between trees prove that a forest works as a cooperative rather than as a competition.

Answer: A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data.

Q30. [BS2-VARC-RC2 · mcsc · diff 0.5]

According to the passage, the fungal threads do all of the following EXCEPT:

- A) gather water and minerals from soil that lies beyond the reach of the roots
- B) produce sugars for themselves rather than take them from the tree's leaves ✓
- C) hold back for themselves a share of the resources that move along them
- D) connect one tree to another when a single fungus is joined to both of them

Answer: B) produce sugars for themselves rather than take them from the tree's leaves

Q31. [BS2-VARC-RC2 · mcsc · diff 0.5]

It can be inferred that the author regards the carbon-tracing experiments as:

- A) flawed in method and therefore in need of repetition by other research teams
- B) sound in what they measure but weak support for the claims built on them ✓
- C) beside the point, since the quantities involved cannot affect survival
- D) conclusive proof that mature trees support shaded seedlings on purpose

Answer: B) sound in what they measure but weak support for the claims built on them

Q32. [BS2-VARC-RC2 · mcsc · diff 0.5]

The comparison with water flowing down a gradient serves mainly to:

- A) show that the fungal threads carry mostly water rather than dissolved sugar
- B) argue that carbon moves too slowly to reach a neighbouring tree in a season
- C) suggest that the labelled carbon was washed through the soil by heavy rain
- D) offer a mechanism for the transfer that needs no intention by either tree ✓

Answer: D) offer a mechanism for the transfer that needs no intention by either tree

Instructions (BS2-VARC-RC3): study the shared passage below and answer its 4 question(s).

Read the following passage carefully and answer the question that follows.

For much of the twentieth century, planners in growing cities treated the bicycle as a relic. Streets were widened, parking was added, and the space left for anyone on two wheels shrank to a painted strip beside fast traffic. Cycling did not disappear, but it became a choice for the brave.

In the last three decades many of those same cities have reversed course. The reason is mostly arithmetic. A lane of road carries far more people per hour on bicycles than in cars, because a bicycle needs about a tenth of the space and, when parked, almost none. Cities that could not afford to widen a single street found they could move more residents simply by changing what the existing width was used for.

Comfort turned out to matter more than exhortation. Where a city merely painted lines on the asphalt, ridership rose a little. Where it separated the cycle lane from moving traffic with a kerb or a row of parked cars, ridership rose sharply, and the new riders included many more children and older people. Surveys give a consistent explanation: most people are willing to cycle but unwilling to cycle beside lorries.

The change has limits. Bicycles work best for trips under about eight kilometres, so a city spread thinly over a wide area will always need other options, and hills and heavy rain reduce the numbers. Freight and emergency vehicles still need road space. What the experience of recent decades suggests is narrower than a slogan: for short journeys in dense districts, the cheapest way to add capacity is often to give a little room to bicycles, and the room only works if it feels safe.

Q33. [BS2-VARC-RC3 · mcsc · diff 0]

Which of the following best expresses the main idea of the passage?

- A) Cities have returned to cycling because residents demanded a healthier way of travelling each day.
- B) Cities have returned to cycling because protected lanes move many people cheaply on short trips. ✓
- C) Cities have returned to cycling because bicycles can replace cars on journeys of almost any length.
- D) Cities have returned to cycling because widening roads for cars proved technically impossible to do.

Answer: B) Cities have returned to cycling because protected lanes move many people cheaply on short trips.

Q34. [BS2-VARC-RC3 · mcsc · diff 0]

The passage mentions all of the following as limits on cycling EXCEPT:

- A) sloping ground and spells of heavy rainfall
- B) the price of buying and keeping up a bicycle ✓
- C) space that freight and emergency vehicles need
- D) journeys longer than roughly eight kilometres

Answer: B) the price of buying and keeping up a bicycle

Q35. [BS2-VARC-RC3 · mcsc · diff 0.5]

It can be inferred that painted cycle lanes attract fewer new riders because:

- A) they are built only in districts where very few residents own a bicycle
- B) they are usually narrower than the width a bicycle actually needs to use
- C) they leave riders beside moving traffic, which most people will not take ✓
- D) they are marked in colours that drivers find difficult to notice in time

Answer: C) they leave riders beside moving traffic, which most people will not take

Q36. [BS2-VARC-RC3 · mcsc · diff 0.5]

The tone of the final paragraph is best described as:

- A) enthusiastic, urging cities to replace their car traffic with bicycles entirely
- B) sceptical, doubting that the recent changes to city streets achieved anything
- C) measured, marking out where cycle lanes help instead of stating a general rule ✓
- D) detached, refusing to take any position at all on the question under review

Answer: C) measured, marking out where cycle lanes help instead of stating a general rule

Q37. [BS2-VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Many plants depend on animals to carry their seeds well away from the parent tree.
- (2) Fleshy fruits attract birds and mammals, which eat the pulp and drop the seeds elsewhere.
- (3) Other seeds carry small hooks that catch on fur and are brushed off much later.
- (4) Wind speed over open grassland is recorded by instruments mounted on tall poles.
- (5) Such journeys reduce the competition between a seedling and the plant that produced it.

Which sentence is the odd one out?

- A) (4) ✓
- B) (2)
- C) (5)
- D) (1)
- E) (3)

Answer: A) (4)

Q38. [BS2-VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Before railways, every town kept its own clock, set roughly by the position of the sun.
- (2) Athletes often train at altitude in the hope of raising the oxygen capacity of their blood.
- (3) Timetables grew confusing once a train could cross many such local times in one day.
- (4) Railway companies therefore imposed a single standard time along the whole of their lines.
- (5) Governments later turned these private arrangements into the time zones still used today.

Which sentence is the odd one out?

- A) (5)
- B) (2) ✓
- C) (4)
- D) (1)
- E) (3)

Answer: B) (2)

Q39. [BS2-VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Trees planted along city streets do far more than soften the look of concrete.
- (2) Their canopies shade the pavement and bring summer temperatures down noticeably.
- (3) Their roots take up rainwater that would otherwise overwhelm drains during a storm.
- (4) Their leaves also trap much of the dust that passing traffic throws into the air.
- (5) Architects still disagree about whether glass facades suit historic city centres.

Which sentence is the odd one out?

- A) (2)
- B) (4)
- C) (5) ✓
- D) (1)
- E) (3)

Answer: C) (5)

Q40. [BS2-VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Riders who still feel unsafe on the road go on avoiding it for months afterwards.
- (2) Planners therefore judge such projects over a span of years rather than weeks.
- (3) When a city paints a cycle lane along a busy road, the effect is rarely immediate.
- (4) Only as the numbers slowly build does the lane start to feel ordinary and safe.

Select the correct order.

- A) 3412
- B) 1432
- C) 3142 ✓
- D) 1342

Answer: C) 3142

Q41. [BS2-VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) After about a week it is lively enough to raise a loaf without any added yeast.
- (2) A sourdough starter is nothing more than flour and water left to attract wild yeasts.
- (3) Bakers who keep feeding it can go on using the same starter for decades.
- (4) Once bubbles appear, the mixture is fed every day with a little fresh flour.

Select the correct order.

- A) 2431
- B) 4231
- C) 2413 ✓
- D) 4213

Answer: C) 2413

Q42. [BS2-VARC-JUMBLE-3 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Rising concern about the emissions of air travel then changed that calculation.
- (2) Cheap flights had made the sleeper look slow and expensive by comparison.
- (3) Several operators have since restored night routes between major European cities.
- (4) Night trains had almost disappeared from Europe by the early years of this century.

Select the correct order.

- A) 2413
- B) 4213 ✓
- C) 4231
- D) 2431

Answer: B) 4213

Q43. [BS2-VARC-JUMBLE-4 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Grams simply remove the guesswork that volume measures quietly introduce.
- (2) Bakers who want repeatable results therefore reach for a kitchen scale instead.
- (3) Recipes in many countries still ask the cook to measure flour by the cupful.
- (4) The same cup, though, holds different weights depending on how tightly it is packed.

Select the correct order.

- A) 4321
- B) 3421 ✓
- C) 3241
- D) 4231

Answer: B) 3421

Q44. [BS2-VARC-SUMMARY-1 · mcsc · diff 0]

Read the paragraph given below and answer the question that follows.

Open-plan offices were introduced partly to encourage conversation between colleagues. In practice many workers find the constant background noise distracting and put on headphones in order to concentrate, which cuts off the very exchanges the layout was meant to create.

Which option best summarises the paragraph?

- A) Workers in open-plan offices wear headphones so that they can concentrate on their own tasks.
- B) Offices ought to be rebuilt with private rooms so that colleagues can talk to one another freely.
- C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off. ✓
- D) Quiet matters more to office productivity than the conversation between colleagues ever did.

Answer: C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off.

Q45. [BS2-VARC-SUMMARY-2 · mcsc · diff 0.5]

Read the paragraph given below and answer the question that follows.

Delivery apps bring a restaurant orders it would never otherwise have received. They also take a share of every bill, and customers who once walked in now stay at home, so a busy kitchen can finish the month having sold more meals and kept less money.

Which option best summarises the paragraph?

- A) Restaurants that use delivery apps must pay a commission on every single order received.
- B) Customers order from home because delivery apps charge less than restaurants themselves do.
- C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each. ✓
- D) Restaurants should abandon delivery apps and serve only customers who come in person.

Answer: C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each.

Q46. [BS2-VARC-SUMMARY-3 · mcsc · diff 0.5]

Read the paragraph given below and answer the question that follows.

Where beavers return to a river they build dams, and parts of the valley flood. Farmers lose the use of some low-lying fields, but the new ponds slow floodwater heading downstream and hold water through dry spells, so the towns below gain what the fields lose.

Which option best summarises the paragraph?

- A) Beavers should be returned to every river, since their dams end both flooding and drought.
- B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security. ✓
- C) Beavers build dams that flood low-lying fields and take useful farmland out of production.
- D) Farmers gain more from beaver ponds than the towns downstream do in long spells of dry weather.

Answer: B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security.

Q47. [BS2-VARC-SUMMARY-4 · mcsc · diff 0.5]

Read the paragraph given below and answer the question that follows.

A translator of poetry cannot keep everything. Sound, rhythm and literal sense pull in different directions, and holding on to one of them usually means loosening the grip on another. Every finished translation is therefore a record of the choices its translator was willing to make.

Which option best summarises the paragraph?

- A) Skilled translators of poetry do manage to preserve sound, rhythm and literal meaning at once.
- B) Poetry is best read in the original, because any translation destroys what really matters in it.
- C) Translators of poetry find it hard to reproduce the rhythm of the original in another language.
- D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts. ✓

Answer: D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts.

Q48. [BS2-VARC-SUMMARY-5 · mcsc · diff 0.5]

Read the paragraph given below and answer the question that follows.

Cities that switch to bright white LED streetlights cut their electricity bills at once. Insects, however, are drawn to that part of the spectrum far more strongly than to the old orange lamps, and the moths gathering under the new lights are eaten or exhausted well before dawn.

Which option best summarises the paragraph?

- A) Moths keep away from white LED lights and so survive better than they did under orange ones.
- B) Cities ought to bring back orange streetlamps in order to protect their declining moths.
- C) The savings from white LED street lighting carry a hidden cost for night-flying insects. ✓
- D) White LED streetlights consume markedly less electricity than the orange lamps they replace.

Answer: C) The savings from white LED street lighting carry a hidden cost for night-flying insects.

Instructions (BS3-VARC-RC1): study the shared passage below and answer its 4 question(s).

For centuries, cartographers left blank the interiors of continents they had not surveyed, and some filled the emptiness with drawings of imagined beasts. These flourishes were not mere decoration. A map is a claim about what is worth recording, and the choice of what to include always reflects the purposes of those who commission it. A trade map emphasises ports, rivers, and the shortest routes between markets; a military map foregrounds ridges, passes, and the width of fords. Neither is a neutral mirror of the land, because no map can show everything at once. To be useful, a map must leave things out.

This selective quality is precisely what makes maps powerful and, at times, misleading. A traveller who trusts a road map assumes that the roads shown are the ones that matter, and rarely wonders about the villages the mapmaker chose to omit. When an authority draws a boundary as a firm line, the line can come to feel as solid as a wall, even where the ground itself offers no such division. In this way a map does not simply describe a territory; it quietly shapes how people imagine and argue about that territory.

Modern digital maps intensify the effect rather than dissolving it. They appear complete and objective, yet every one rests on decisions about which streets to label, which businesses to display, and which neighbourhoods to render in detail. The reader sees a clean surface and forgets the editing beneath it. Understanding a map, then, requires more than reading its symbols. It requires asking who made it, for whom, and what they had reason to leave in the blank spaces.

Q49. [BS3-VARC-RC1 · mcsc · diff 0.5]

Which of the following best expresses the main idea of the passage?

- A) Trade maps and military maps differ chiefly because rivers matter more to merchants than mountain passes ever will.
- B) Early cartographers decorated unknown regions with imaginary creatures to disguise the limits of their knowledge from buyers.
- C) Digital maps have finally removed the distortions that troubled the printed maps of earlier centuries and travellers.
- D) Maps are selective representations whose contents reflect the aims of their makers rather than a neutral picture of the land. ✓

Answer: D) Maps are selective representations whose contents reflect the aims of their makers rather than a neutral picture of the land.

Solution: The passage repeatedly argues that maps leave things out according to the maker's purpose and thus are not neutral mirrors, which the selective-representation option captures. The creatures point is a single opening example, not the thesis. The claim that digital maps remove distortion is contradicted, since the text says they intensify the effect. The trade-versus-military contrast is one illustration, not the overall point.

Q50. [BS3-VARC-RC1 · mcsc · diff 0.5]

According to the passage, all of the following are true of maps EXCEPT:

- A) The features a map emphasises depend on the purpose for which it was commissioned by its makers.
- B) A digital map presents a complete and objective record precisely because it labels every street and business. ✓
- C) A boundary drawn as a firm line can come to feel solid even where the land shows no real division.
- D) A map cannot display everything at once, so being useful depends on deliberately leaving certain things out.

Answer: B) A digital map presents a complete and objective record precisely because it labels every street and business.

Solution: The passage says digital maps only appear complete and objective while resting on hidden editing choices, so treating them as genuinely complete is the claim the text denies. The other statements are directly supported: maps must omit to be useful, a firm boundary line can feel like a wall, and emphasis follows the commissioning purpose.

Q51. [BS3-VARC-RC1 · mcsc · diff 0.5]

It can be inferred from the passage that a reader who wants to judge a map fairly should:

- A) assume that the roads a road map displays are always the ones that truly matter
- B) prefer digital maps to printed ones because their surfaces are cleaner and easier
- C) consider the intentions behind the map and notice what its maker chose not to show ✓
- D) accept that any boundary drawn as a firm line marks a genuine division in the land

Answer: C) consider the intentions behind the map and notice what its maker chose not to show

Solution: The closing lines say understanding a map requires asking who made it, for whom, and what was left in the blank spaces, so a fair reader weighs purpose and omission. Preferring digital maps ignores the warning that clean surfaces hide editing. Trusting firm boundaries and displayed roads repeats the very errors the passage cautions against.

Q52. [BS3-VARC-RC1 · mcsc · diff 0.5]

Why does the author mention trade maps and military maps in the first paragraph?

- A) To suggest that the two kinds of map were usually produced by the same workshops
- B) To prove that military needs have always mattered far more than trade did to early states everywhere
- C) To show that different aims lead makers to highlight different features of the same land ✓
- D) To argue that rivers are harder to chart accurately than ridges and mountain passes

Answer: C) To show that different aims lead makers to highlight different features of the same land

Solution: The two examples illustrate the point that a map's emphasis follows its purpose, one stressing routes to markets and the other stressing terrain for movement. The passage makes no ranking of military over trade, offers no claim about charting difficulty, and says nothing about shared workshops.

Instructions (BS3-VARC-RC2): study the shared passage below and answer its 4 question(s).

When a forest burns, the loss looks total. Yet many woodlands have evolved not merely to survive fire but to depend on it. The cones of certain pines are sealed shut with resin and open only when the heat of a blaze melts that seal, releasing seeds onto ground newly cleared of competitors and enriched with ash. For such trees, a fire is not a disaster but the signal to reproduce. Ecologists describe these systems as fire-adapted, and they are far more common than the popular image of fire as pure destruction suggests.

The difficulty is that not all fires are alike. A frequent, low-intensity fire clears undergrowth while leaving mature trees standing, and many species are tuned to exactly this rhythm. When people suppress every small fire in order to protect property, dead wood and dry brush accumulate year after year. The next fire, when it finally comes, burns hotter and higher than anything the forest is built to withstand, killing even the trees that would ordinarily endure. In trying to prevent fire altogether, a policy can make the eventual blaze far more severe.

This paradox has forced a rethinking of how forests are managed. In several regions, foresters now set controlled burns on purpose, choosing cool, damp days when a fire can be kept small and contained. The aim is not to eliminate fire but to restore its natural frequency, so that fuel never builds to dangerous levels. The approach demands patience and public trust, since a deliberate fire is an unsettling sight. But it reflects a hard lesson: a landscape shaped by fire cannot be kept safe by pretending fire away.

Q53. [BS3-VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Foresters in several regions have recently begun to distrust the public's reaction to any deliberate fire.
- B) Because many forests are adapted to regular fire, suppressing all fires can make later blazes more destructive. ✓
- C) The resin-sealed cones of certain pines will open only when a passing blaze first melts the seal and then frees the seeds.
- D) Controlled burns should be set on damp days so that the flames stay small, contained, and easy to watch.

Answer: B) Because many forests are adapted to regular fire, suppressing all fires can make later blazes more destructive.

Solution: The passage builds from fire-adapted forests to the paradox that blanket suppression lets fuel accumulate and produces worse fires, which the correct option summarises. The damp-day detail and the pine cones are supporting examples, and public distrust is a practical hurdle mentioned in passing, not the core claim.

Q54. [BS3-VARC-RC2 · mcsc · diff 0.5]

According to the passage, why does suppressing every small fire eventually lead to a more severe blaze?

- A) The resin seals on pine cones grow thicker each season and demand a hotter fire to open.
- B) Dead wood and dry brush pile up over the years, giving a later fire far more fuel to burn. ✓
- C) Mature trees lose their natural resistance to heat when they are protected for too long.
- D) Damp weather becomes rarer over time, so foresters can no longer schedule any cool burns.

Answer: B) Dead wood and dry brush pile up over the years, giving a later fire far more fuel to burn.

Solution: The second paragraph states that when small fires are suppressed, dead wood and dry brush accumulate year after year, so the next fire burns hotter and higher. The other options invent mechanisms the passage never gives: thickening seals, lost heat resistance, or vanishing damp weather.

Q55. [BS3-VARC-RC2 · msc · diff 0.5]

It can be inferred that the author would most likely agree with which of the following statements?

- A) Popular opinion about fire is usually a reliable guide to how forests should actually be managed.
- B) The safest forest is one in which every fire, however small, is put out as quickly as it can be found.
- C) Protecting the property of nearby residents should never once be counted among the goals of any modern forest-management plan.
- D) A management policy works better when it cooperates with a landscape's natural patterns than when it fights them. ✓

Answer: D) A management policy works better when it cooperates with a landscape's natural patterns than when it fights them.

Solution: The passage praises restoring fire's natural frequency and warns against pretending fire away, so the author favours working with natural patterns. Dismissing property protection goes further than the text, which treats it as a real concern. Extinguishing every fire is the very policy criticised, and the passage says the popular image of fire is misleading.

Q56. [BS3-VARC-RC2 · msc · diff 0.5]

Which word best describes the author's overall tone in the passage?

- A) Measured and explanatory, presenting a corrective view of a common belief ✓
- B) Alarmed and urgent, warning that forests everywhere face imminent collapse
- C) Playful and ironic, mocking the foresters who still set deliberate burns
- D) Nostalgic and mournful, lamenting woodlands that can no longer be saved

Answer: A) Measured and explanatory, presenting a corrective view of a common belief

Solution: The author calmly explains fire adaptation and the suppression paradox to correct the popular image of fire as pure destruction, a measured and explanatory stance. The tone is not alarmed about imminent collapse, not mournful about lost forests, and not mocking of foresters, whose controlled-burn practice the passage endorses.

Instructions (BS3-VARC-RC3): study the shared passage below and answer its 4 question(s).

The public library began as a simple idea: that useful books should be available to anyone, not only to those who could afford to buy them. Early lending libraries charged a small fee, which kept out the very readers who might have gained the most. The breakthrough came when towns agreed to pay for libraries out of shared taxes, so that borrowing became free at the point of use. A reader needed only to live in the town and to return the books on time. For many families, the library was the first place where knowledge cost nothing.

Over time, libraries grew into something larger than shelves of books. They became quiet places to study, warm rooms to sit in, and centres where people could ask a trained librarian for help finding what they needed. Children attended reading hours, job seekers used the desks, and newcomers to a town found notices about local life. The building offered not just texts but a shared space open to all, regardless of wealth.

Today, when a great deal of information sits a click away, some people ask whether libraries are still needed. Yet the answer many communities give is yes. Not everyone owns a computer or a quiet place to work, and not every question is easily answered by a search engine. The library remains what it has long been: a place where help and knowledge are offered freely, to whoever walks through the door.

Q57. [BS3-VARC-RC3 · msc · diff 0]

Which of the following best captures the main point of the passage?

- A) Librarians spend more of their time helping job seekers than assisting young children.
- B) The public library has long offered free access to knowledge and remains valuable today. ✓
- C) Early lending libraries failed mainly because the fees they charged were set far too high.
- D) Search engines have replaced most of the services that public libraries once provided.

Answer: B) The public library has long offered free access to knowledge and remains valuable today.

Solution: The passage traces the library from free tax-funded lending to a broad community space and argues it is still needed, which the correct option captures. The fee point is background, the search-engine claim is the view the passage rejects, and the comparison of job seekers to children is never made.

Q58. [BS3-VARC-RC3 · mcsc · diff 0]

According to the passage, what change made borrowing free at the point of use?

- A) Librarians volunteered their time so that no salaries had to be paid at all.
- B) Towns agreed to fund libraries through shared taxes instead of charging each reader. ✓
- C) Publishers lowered the price of books until borrowing them cost almost nothing.
- D) Wealthy donors bought up large private collections of books and then handed them over to the towns.

Answer: B) Towns agreed to fund libraries through shared taxes instead of charging each reader.

Solution: The first paragraph says the breakthrough came when towns agreed to pay for libraries out of shared taxes, making borrowing free at the point of use. Donors, volunteer librarians, and cheaper book prices are not mentioned as the cause.

Q59. [BS3-VARC-RC3 · mcsc · diff 0.5]

It can be inferred from the passage that the author regards the claim that libraries are no longer needed as:

- A) reasonable, since a search engine now answers nearly every possible question
- B) unimportant, because libraries were never truly meant to serve the whole public
- C) mistaken, because many people still lack the resources a library freely provides ✓
- D) proven, given that most families already own computers of their own at home

Answer: C) mistaken, because many people still lack the resources a library freely provides

Solution: The final paragraph answers the doubt with a yes, noting that not everyone owns a computer or a quiet workspace and that search engines cannot answer every question, so the author treats the claim as mistaken. The other options either accept the doubt or contradict the passage's insistence that the library serves everyone.

Q60. [BS3-VARC-RC3 · mcsc · diff 0.5]

What is the main purpose of the second paragraph?

- A) To compare the reading hours for children with the desks used by job seekers
- B) To show that libraries came to serve as shared community spaces, not only stores of books ✓
- C) To describe how librarians are trained before they begin helping the public
- D) To explain the exact set of rules that a reader had to carefully follow in order to borrow any book

Answer: B) To show that libraries came to serve as shared community spaces, not only stores of books

Solution: The second paragraph lists study space, warmth, librarian help, reading hours, and notices to show the library grew into a shared community space beyond its shelves. It does not lay out borrowing rules, describe librarian training, or set up a comparison between children and job seekers.

Q61. [BS3-VARC-ODD-1 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) Honeybees communicate the location of food through a series of movements known as the waggle dance.
- (2) The angle of the dance relative to vertical tells other bees the direction of the flowers.
- (3) The duration of each waggle run signals roughly how far away the food source lies.
- (4) A large hive can hold tens of thousands of individual worker bees during summer.
- (5) Observers can even estimate distance by timing how long each run lasts.

Which sentence is the odd one out?

- A) Sentence 1
- B) Sentence 4 ✓
- C) Sentence 3
- D) Sentence 5
- E) Sentence 2

Answer: B) Sentence 4

Solution: Sentences 1, 2, 3 and 5 all describe how the waggle dance encodes the direction and distance of food, forming one connected idea. Sentence 4 shifts to hive population, an unrelated statistic that breaks the theme, so it is the misfit. The other options each belong to the dance-communication chain.

Q62. [BS3-VARC-ODD-2 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) The invention of movable type allowed identical pages to be printed quickly and cheaply.
- (2) Books that once took months to copy by hand could suddenly be produced in days.
- (3) Wider access to printed texts helped standardise spelling across many regions.
- (4) Lower prices meant that ideas spread far beyond wealthy monasteries and courts.
- (5) Many early manuscripts were decorated with gold leaf and elaborate hand-painted borders.

Which sentence is the odd one out?

- A) Sentence 2
- B) Sentence 3
- C) Sentence 5 ✓
- D) Sentence 1
- E) Sentence 4

Answer: C) Sentence 5

Solution: Sentences 1, 2, 3 and 4 all trace the effects of printing, faster output, standard spelling and cheaper wider access, so they cohere. Sentence 5 describes the decoration of hand-made manuscripts, which runs against the theme of mass printing and is the odd one out. The remaining options each extend the printing-impact idea.

Q63. [BS3-VARC-ODD-3 · mcsc · diff 0.5]

Five sentences are given below. Four of them can be joined to form a coherent paragraph on a single idea, while one does not fit.

- (1) Coral reefs support a quarter of all marine species despite covering a tiny fraction of the ocean.
- (2) The corals depend on algae living inside their tissues to provide most of their food.
- (3) When water grows too warm, the corals expel these algae and turn a pale white.
- (4) A prolonged loss of algae can leave the corals starving and eventually dead.
- (5) Deep-sea trenches remain among the least explored regions of the entire planet.

Which sentence is the odd one out?

- A) Sentence 5 ✓
- B) Sentence 3
- C) Sentence 1
- D) Sentence 4
- E) Sentence 2

Answer: A) Sentence 5

Solution: Sentences 1 to 4 build one picture of coral reefs and their reliance on algae, including the bleaching that follows warming. Sentence 5 jumps to deep-sea trenches, a separate ocean topic that does not connect, making it the misfit. Each other option contributes to the coral-and-algae narrative.

Q64. [BS3-VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. A city planted thousands of trees along its hottest streets to cool them down.
2. Within a few summers, shaded pavements were measurably cooler than bare ones.
3. Residents also reported walking more often once the routes felt pleasant.
4. The idea began as a simple response to rising temperatures in the centre.

Select the correct order.

- A) 4-1-2-3 ✓
- B) 1-2-3-4
- C) 4-2-1-3
- D) 1-4-3-2

Answer: A) 4-1-2-3

Solution: Sentence 4 opens with the origin of the idea, sentence 1 states the action taken, sentence 2 gives the first measured result, and sentence 3 adds a further outcome. The order 4-1-2-3 moves from motive to action to results. Sequences starting with 1 skip the framing motive, and 4-2-1-3 places a result before the action that caused it.

Q65. [BS3-VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. This layer traps heat and keeps the surface warm enough for liquid water.
2. The atmosphere of a planet is the blanket of gases held by its gravity.
3. Without it, the same surface would swing between scorching days and freezing nights.
4. On our world that blanket is mostly nitrogen with a smaller share of oxygen.

Select the correct order.

- A) 2-4-1-3 ✓
- B) 2-1-4-3
- C) 4-2-1-3
- D) 2-3-1-4

Answer: A) 2-4-1-3

Solution: Sentence 2 defines the atmosphere, sentence 4 names its main gases, sentence 1 explains the warming effect of that layer, and sentence 3 states the consequence of its absence. The order 2-4-1-3 flows from definition to composition to function to contrast. Options that put 1 before 4 mention the layer before it is described, and 4 cannot open since it refers back to a blanket not yet named.

Q66. [BS3-VARC-JUMBLE-3 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. She began by sorting the beans by size before the roasting could start.
2. A small cafe decided to roast its own coffee for the first time.
3. Only then did the owner adjust the heat to bring out the flavour.
4. Uneven beans, she knew, would scorch and cook at different rates.

Select the correct order.

- A) 2-1-4-3 ✓
- B) 2-3-1-4
- C) 1-4-2-3
- D) 2-4-1-3

Answer: A) 2-1-4-3

Solution: Sentence 2 sets the scene, sentence 1 describes the first step of sorting, sentence 4 explains why that step matters, and sentence 3 uses 'only then' to mark the next stage. The order 2-1-4-3 keeps action before its justification and reserves 3 for last. Placing 4 before 1 explains a step not yet mentioned, and 3 cannot precede the sorting it depends on.

Q67. [BS3-VARC-JUMBLE-4 · mcsc · diff 0.5]

The four sentences below, labelled 1 to 4, form a paragraph when placed in the correct order. Rearrange them into the most logical sequence.

1. Early maps of the region were drawn mostly from travellers' rough sketches.
2. As a result, coastlines were often placed dozens of miles from their true position.
3. Later surveys with proper instruments corrected many of these old errors.
4. Today satellites fix the same coastlines to within a few metres.

Select the correct order.

- A) 3-1-2-4
- B) 1-3-2-4
- C) 1-2-4-3
- D) 1-2-3-4 ✓

Answer: D) 1-2-3-4

Solution: The paragraph is chronological: sentence 1 describes early sketch-based maps, sentence 2 gives their result, sentence 3 introduces later instrument surveys, and sentence 4 ends with modern satellites. The order 1-2-3-4 follows time and cause. Swapping 2 and 3 separates the error from its cause, and ending on 3 rather than 4 reverses the march toward greater accuracy.

Q68. [BS3-VARC-SUMMARY-1 · mcsc · diff 0]

Libraries once measured their worth by how many books they owned, but many now judge themselves by how well they connect people to information wherever it lives. A modern library lends digital files, offers work and meeting space, and guides visitors through vast online collections it does not physically hold. The building still matters, yet its main job has shifted from storing objects to giving access.

Which option best summarises the paragraph?

- A) Libraries now judge their worth by how well they connect people to information rather than by how many books they own. ✓
- B) Modern libraries lend out digital files and provide quiet meeting rooms as well as long shelves of printed books for their many visitors.
- C) Libraries no longer need any physical building at all, since nearly every part of their collection is now stored and read online.
- D) The sheer number of books that a library owns remains the clearest and most reliable measure of how useful it truly is.

Answer: A) Libraries now judge their worth by how well they connect people to information rather than by how many books they own.

Solution: The passage contrasts an older ownership measure with a new access measure, which the correct option captures fully. The digital-and-meeting-room option is too narrow, listing services without the central shift. Claiming buildings are unneeded overreaches beyond the text, which says the building still matters, and the counting-books option restates the old view the passage rejects.

Q69. [BS3-VARC-SUMMARY-2 · mcsc · diff 0.5]

Sleep is often treated as lost time, yet the brain stays busy during it. While a person rests, the brain sorts through the day's experiences, strengthens useful memories and clears away waste that builds up during waking hours. Far from being idle, sleep is active maintenance that keeps the mind working well the next day.

Which option best summarises the paragraph?

- A) Sleep feels like wasted time to most people because nothing worth measuring ever happens inside the resting brain.
- B) People who sleep for longer will always think faster and remember far more facts than those who choose to sleep much less.
- C) During each period of sleep the brain quietly strengthens the useful memories a person has formed across the whole of the previous day.
- D) Sleep is not lost time but active brain upkeep that sorts memories and clears waste to keep the mind sharp. ✓

Answer: D) Sleep is not lost time but active brain upkeep that sorts memories and clears waste to keep the mind sharp.

Solution: The main point is that sleep is productive upkeep involving memory sorting and waste clearing, which the correct option states in full. The memory-strengthening option is too narrow, covering only one function. The 'always faster' option overreaches into a claim the passage never makes, and the 'nothing happens' option distorts the text, which argues the opposite.

Q70. [BS3-VARC-SUMMARY-3 · mcsc · diff 0.5]

Fixing a bicycle taught the student more than any lecture had. By taking the gears apart and putting them back together, she saw how each part depended on the others in a way diagrams had never made clear. The lesson was not really about bicycles; it was that handling something directly can reveal what mere description leaves hidden.

Which option best summarises the paragraph?

- A) Bicycles are surprisingly complicated machines whose many gears turn out to depend on one another in unexpected ways.
- B) The student slowly learned how to take the gears of a bicycle apart and then reassemble them correctly during her afternoon repair.
- C) Handling a real object directly can reveal how its parts relate in ways that descriptions and diagrams cannot. ✓
- D) Lectures and diagrams are quite useless for teaching almost anything and should be replaced entirely by hands-on practice sessions.

Answer: C) Handling a real object directly can reveal how its parts relate in ways that descriptions and diagrams cannot.

Solution: The paragraph generalises from one repair to a claim about hands-on learning, which the correct option captures. The gear-reassembly option is too narrow, sticking to the literal task. The 'lectures are useless' option overreaches, since the text only says direct handling reveals more, and the bicycle-complexity option treats the example as the point rather than the lesson about learning.

Q71. [BS3-VARC-SUMMARY-4 · mcsc · diff 0.5]

A town worried about litter tried fines for a year with little effect. It then placed more bins and painted cheerful footpaths leading to them, and litter fell sharply. The change suggested that making the right action easy and inviting often works better than simply punishing the wrong one.

Which option best summarises the paragraph?

- A) Fines can never change the way the public behaves and ought to be abolished by every town that seriously hopes to stay clean.
- B) Making a desired action easy and inviting can cut a problem more than punishing the unwanted behaviour does. ✓
- C) The town first tried out fines for a full year before it finally decided to add many more bins around its busy streets.
- D) The town carefully painted a set of cheerful footpaths that were meant to guide residents toward its newly installed litter bins.

Answer: B) Making a desired action easy and inviting can cut a problem more than punishing the unwanted behaviour does.

Solution: The passage draws a general lesson that easing the right action beats punishing the wrong one, which the correct option states. The painted-footpaths option is too narrow, naming one tactic only. The 'fines never work' option overreaches beyond the single case, and the fines-first-for-a-year option reports a detail without the main conclusion.

Q72. [BS3-VARC-SUMMARY-5 · mcsc · diff 0.5]

Maps always leave things out, and that is their strength. A useful subway map ignores true distances and street layouts so that riders can see the lines and stops at a glance. By dropping detail that would only clutter the task at hand, a good map makes the information that matters easy to read.

Which option best summarises the paragraph?

- A) Maps that leave out any information are basically misleading and cannot really be trusted for any kind of serious purpose.
- B) Subway riders simply need to see the coloured lines and the stops so that they can plan their daily journeys across the city.
- C) Subway maps very deliberately ignore both the true distances between stations and the real layout of the streets above them.
- D) A good map is useful because it leaves out detail that would clutter the task, showing only what matters. ✓

Answer: D) A good map is useful because it leaves out detail that would clutter the task, showing only what matters.

Solution: The central claim is that leaving out detail is what makes a map useful, which the correct option expresses. The subway-distances option is too narrow, giving only the example. The 'maps are misleading' option distorts the passage, which praises omission, and the riders-plan-journeys option states a side detail rather than the main idea.

Section: DILR (40 min)

Instructions (BS1-DILR-1): study the shared data below and answer its 3 question(s).

Four teams, Alfa, Bravo, Comet and Delta, play a single round-robin tournament: every team plays every other team exactly once (6 matches in all). A win is worth 3 points, a tie is worth 1 point to each team, and a loss is worth 0 points. The following facts are known about the completed tournament:

- Alfa won every match it played. - Delta lost every match it played. - Exactly one match in the whole tournament ended in a tie. - That single tie was the match between Bravo and Comet.

Q73. [BS1-DILR-1 · mcsc · diff 0.5]

How many points did Comet earn in the tournament?

A) 4 points ✓

B) 7 points

C) 3 points

D) 6 points

Answer: A) 4 points

Q74. [BS1-DILR-1 · mcsc · diff 0.5]

How many of the 6 matches ended with a winner (that is, did NOT end in a tie)?

A) 6 matches

B) 4 matches

C) 5 matches ✓

D) 3 matches

Answer: C) 5 matches

Q75. [BS1-DILR-1 · mcsc · diff 0.5]

Which statement about the final points table is correct?

A) Comet finished ahead of Bravo on points, with Alfa on top.

B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta. ✓

C) Delta finished level on points with Bravo, behind Alfa and Comet.

D) Alfa and Bravo finished level on points at the top, ahead of Comet and Delta.

Answer: B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta.

Instructions (BS1-DILR-2): study the shared data below and answer its 3 question(s).

Three friends, Mira, Nina and Owen, each take the same short quiz of 5 questions. Every correct answer is worth 3 points; a wrong or blank answer scores 0 points. So each person's score equals 3 times the number of questions they answered correctly. The following facts are known:

- Between them, the three friends answered exactly 9 questions correctly. - Mira answered more questions correctly than Nina, and Nina answered more correctly than Owen. - Owen answered exactly 2 questions correctly. - No one answered all 5 questions correctly.

Q76. [BS1-DILR-2 · mcsc · diff 0.5]

How many points did Mira score?

A) 9 points

B) 6 points

C) 12 points ✓

D) 15 points

Answer: C) 12 points

Q77. [BS1-DILR-2 · mcsc · diff 0.5]

How many questions did Nina answer correctly?

A) 4 questions

B) 3 questions ✓

C) 2 questions

D) 5 questions

Answer: B) 3 questions

Q78. [BS1-DILR-2 · mcsc · diff 0.5]

What was the combined total score of all three friends?

- A) 27 points ✓
- B) 30 points
- C) 9 points
- D) 18 points

Answer: A) 27 points

Instructions (BS1-DILR-3): study the shared data below and answer its 4 question(s).

Units sold during the month (number of packets)

Product	Aria	Bani	Cera	Dovi
Tea	120	200	150	180
Coffee	80	150	200	120
Cocoa	60	100	50	90

Selling price per packet (in rupees)

Product	Price
Tea	20
Coffee	30
Cocoa	40

Q79. [BS1-DILR-3 · mcsc · diff 0.5]

How many packets in total did store Bani sell across all three products during the month?

- A) 350
- B) 400
- C) 450 ✓
- D) 260

Answer: C) 450

Q80. [BS1-DILR-3 · mcsc · diff 0.5]

What was the total sales revenue (in rupees) earned by store Cera from all three products?

- A) 12500
- B) 11000 ✓
- C) 10800
- D) 7200

Answer: B) 11000

Q81. [BS1-DILR-3 · mcsc · diff 0.5]

What was the combined revenue (in rupees) from Tea across all four stores?

- A) 650
- B) 16500
- C) 11000
- D) 13000 ✓

Answer: D) 13000

Q82. [BS1-DILR-3 · mcsc · diff 0.5]

In how many of the four stores did the total number of packets sold (all products combined) exceed 350?

- A) 4
- B) 3 ✓
- C) 1
- D) 2

Answer: B) 3

Instructions (BS1-DILR-4): study the shared data below and answer its 3 question(s).

Three investors, Priya, Rohan and Sara, each put a whole number of lakhs of rupees into a separate fund at the start of the year. The following facts are known:

- Together the three invested a total of 24 lakh. - Rohan invested exactly twice as much as Priya. - Sara invested 4 lakh more than Priya.

Over the year the funds gave these returns on the amount invested: Priya's fund returned 20%, Rohan's fund returned 10%, and Sara's fund returned 30%.

Q83. [BS1-DILR-4 · mcsc · diff 0.5]

How much did Rohan invest?

- A) 5 lakh
- B) 10 lakh ✓
- C) 9 lakh
- D) 12 lakh

Answer: B) 10 lakh

Q84. [BS1-DILR-4 · mcsc · diff 0.5]

What was the combined amount invested by Priya and Sara together?

- A) 19 lakh
- B) 10 lakh
- C) 15 lakh
- D) 14 lakh ✓

Answer: D) 14 lakh

Q85. [BS1-DILR-4 · mcsc · diff 1]

After the year's returns are added to the amounts invested, which investor ends with the largest total amount of money?

- A) Rohan and Sara end level, each with 11 lakh
- B) Sara, with 11.7 lakh ✓
- C) Rohan, with 11 lakh
- D) Priya, with 6 lakh

Answer: B) Sara, with 11.7 lakh

Instructions (BS1-DILR-5): study the shared data below and answer its 5 question(s).

Units produced by a factory in each quarter

Product	Q1	Q2	Q3
Steel	400	500	600
Cement	300	300	450
Glass	200	250	200
Wood	100	150	250

Production cost per unit (in rupees)

Product	Rate
Steel	50
Cement	40
Glass	60
Wood	30

Q86. [BS1-DILR-5 · mcsc · diff 0.5]

What was the total number of Steel units produced across the three quarters?

- A) 1500 ✓
- B) 1050
- C) 1100
- D) 1300

Answer: A) 1500

Q87. [BS1-DILR-5 · mcsc · diff 0.5]

By what percentage did Steel production increase from Q1 to Q3?

- A) 20
- B) 200
- C) 33
- D) 50 ✓

Answer: D) 50

Q88. [BS1-DILR-5 · mcsc · diff 0.5]

What was the average quarterly production of Cement over the three quarters?

- A) 300
- B) 1050
- C) 350 ✓
- D) 375

Answer: C) 350

Q89. [BS1-DILR-5 · mcsc · diff 0.5]

What is the largest single value of units produced by any one product in any one quarter?

- A) 500
- B) 450
- C) 250
- D) 600 ✓

Answer: D) 600

Q90. [BS1-DILR-5 · mcsc · diff 0.5]

What was the total production cost (in rupees) of all four products in Q1?

- A) 47000 ✓
- B) 44000
- C) 1000
- D) 50000

Answer: A) 47000

Instructions (BS2-DILR-1): study the shared data below and answer its 3 question(s).

****Read the information given below and answer the question that follows.****

Five runners, Farid, Lena, Mirko, Priya and Tomas, finished a 5 km club race in five different positions, from first to fifth. No two runners finished together.

- Priya finished ahead of Lena, and Lena finished ahead of Mirko. - Exactly one runner finished between Farid and Tomas, with Farid ahead of Tomas. - Priya did not finish first.

Q91. [BS2-DILR-1 · mcsc · diff 0.5]

Which runner finished third?

- A) Priya
- B) Tomas ✓
- C) Farid
- D) Lena

Answer: B) Tomas

Q92. [BS2-DILR-1 · mcsc · diff 0.5]

In which position did Lena finish?

- A) Fourth ✓
- B) Second
- C) Fifth
- D) Third

Answer: A) Fourth

Q93. [BS2-DILR-1 · mcsc · diff 0.5]

Which of the following statements is true?

- A) Lena finished immediately behind Tomas but ahead of Priya
- B) Tomas finished immediately ahead of Priya
- C) Mirko finished immediately behind Lena ✓
- D) Farid finished two places behind Priya

Answer: C) Mirko finished immediately behind Lena

Instructions (BS2-DILR-2): study the shared data below and answer its 3 question(s).

****Read the information given below and answer the question that follows.****

Four members of a college club, Anya, Boris, Carla and Dev, earned points in a month-long fundraising drive. Together they earned 120 points.

- Boris earned 6 points more than Carla. - Anya earned exactly twice as many points as Dev. - Dev earned 16 points.

Q94. [BS2-DILR-2 · mcsc · diff 0.5]

How many points did Carla earn?

- A) 36
- B) 33 ✓
- C) 30
- D) 39

Answer: B) 33

Q95. [BS2-DILR-2 · mcsc · diff 0.5]

Who earned the most points?

- A) Carla
- B) Dev
- C) Boris ✓
- D) Anya

Answer: C) Boris

Q96. [BS2-DILR-2 · mcsc · diff 0.5]

By how many points does the highest total exceed the lowest total?

- A) 23 ✓
- B) 27
- C) 7
- D) 17

Answer: A) 23

Instructions (BS2-DILR-3): study the shared data below and answer its 4 question(s).

****Study the data given below and answer the question that follows.****

A beverage wholesaler reports last month's trade at its three depots. Table 1 gives the number of packs sold, in hundreds of packs. Table 2 gives the selling price of one pack, in euros, at that depot.

Table 1: Packs sold (in hundreds)

Category	Verona	Padua	Ferrara
Espresso blend	12	15	9
Herbal tea	8	6	10
Cocoa	5	9	11

Table 2: Price per pack (in euros)

Category	Verona	Padua	Ferrara
Espresso blend	20	20	25
Herbal tea	10	15	10
Cocoa	30	20	20

Q97. [BS2-DILR-3 · mcsc · diff 0.5]

How many hundreds of packs were sold in total at the Ferrara depot?

- A) 30 ✓
- B) 28
- C) 25
- D) 32

Answer: A) 30

Q98. [BS2-DILR-3 · mcsc · diff 0.5]

Which depot and category together generated the highest revenue?

- A) Padua, espresso blend ✓
- B) Ferrara, cocoa
- C) Verona, espresso blend
- D) Ferrara, espresso blend

Answer: A) Padua, espresso blend

Q99. [BS2-DILR-3 · mcsc · diff 0.5]

What was the combined revenue from cocoa across the three depots, in euros?

- A) 52,000
- B) 61,000
- C) 55,000 ✓
- D) 58,000

Answer: C) 55,000

Q100. [BS2-DILR-3 · mcsc · diff 0.5]

In how many of the nine depot and category combinations did revenue exceed 20,000 euros?

- A) 3
- B) 4 ✓
- C) 6
- D) 5

Answer: B) 4

Instructions (BS2-DILR-4): study the shared data below and answer its 3 question(s).

****Read the information given below and answer the question that follows.****

Four volunteers, Ines, Karim, Nadia and Omar, each work exactly one lab shift during a week, on Monday, Tuesday, Wednesday and Thursday, one volunteer per day. Each is also assigned a different single duty: filing, inventory, reception or training.

- Karim works on the day immediately after Nadia's shift. - Ines works neither on Monday nor on Thursday. - Omar's shift falls earlier in the week than Ines's shift. - The volunteer working on Wednesday is assigned inventory. - Ines is assigned reception, and Omar is assigned training.

Q101. [BS2-DILR-4 · mcsc · diff 0.5]

Which volunteer works on Monday?

- A) Omar ✓
- B) Karim
- C) Ines
- D) Nadia

Answer: A) Omar

Q102. [BS2-DILR-4 · mcsc · diff 0.5]

Which volunteer and duty are correctly paired?

- A) Omar with reception
- B) Karim with inventory
- C) Nadia with filing
- D) Karim with filing ✓

Answer: D) Karim with filing

Q103. [BS2-DILR-4 · mcsc · diff 1]

Suppose Ines and Karim exchange shift days, every duty staying with the same person. Who then works on the day immediately before the inventory shift?

- A) Karim ✓
- B) Nadia
- C) Omar
- D) Ines

Answer: A) Karim

Instructions (BS2-DILR-5): study the shared data below and answer its 5 question(s).

****Study the data given below and answer the question that follows.****

The bar chart shows the number of bicycles sold, in thousands, for three models of a manufacturer in each of the years 2021, 2022 and 2023.

Q104. [BS2-DILR-5 · mcsc · diff 0.5]

How many thousand bicycles were sold in total across the three models in 2022?

- A) 63 ✓
- B) 65
- C) 60
- D) 58

Answer: A) 63

Q105. [BS2-DILR-5 · mcsc · diff 0.5]

By what percentage did Kestrel sales grow from 2021 to 2023?

- A) 120%
- B) 140% ✓
- C) 58%
- D) 240%

Answer: B) 140%

Q106. [BS2-DILR-5 · mcsc · diff 0.5]

What is the ratio of total Orion sales over the three years to total Nimbus sales over the three years?

- A) 3 : 5
- B) 5 : 4
- C) 4 : 5 ✓
- D) 2 : 3

Answer: C) 4 : 5

Q107. [BS2-DILR-5 · mcsc · diff 0.5]

What were the average annual sales of Kestrel, in thousands, over the three years?

- A) 20
- B) 18 ✓
- C) 17
- D) 16

Answer: B) 18

Q108. [BS2-DILR-5 · mcsc · diff 0.5]

By how many thousand units did the three-model total in 2023 exceed the three-model total in 2021?

- A) 34
- B) 40
- C) 38
- D) 36 ✓

Answer: D) 36

Instructions (BS3-DILR-1): study the shared data below and answer its 3 question(s).

Five runners, Neha, Omar, Priya, Rahul and Sana, finished a road race in positions one (first) through five (last), with no ties.

- Omar finished last. - Priya finished immediately before Rahul. - Neha finished ahead of both Priya and Sana. - Sana did not finish fourth.

Q109. [BS3-DILR-1 · mcsc · diff 0.5]

Who finished in second place?

- A) Sana ✓
- B) Priya
- C) Neha
- D) Rahul

Answer: A) Sana

Solution: Omar is fifth. The Priya-Rahul block sits at positions three and four, so Neha and Sana take positions one and two. Since Neha finishes ahead of Sana, Neha is first and Sana is second. Neha is the wrong pick from reading the block, while Priya and Rahul are fixed at third and fourth.

Q110. [BS3-DILR-1 · mcsc · diff 0.5]

Who finished in fourth place?

- A) Priya
- B) Sana
- C) Rahul ✓
- D) Omar

Answer: C) Rahul

Solution: The Priya-Rahul block occupies positions three and four, placing Rahul fourth. Priya is one place earlier at third, Sana is second, and Omar is fixed last rather than fourth.

Q111. [BS3-DILR-1 · mcsc · diff 0.5]

How many runners finished between Sana and Omar?

- A) Three
- B) Zero
- C) One
- D) Two ✓

Answer: D) Two

Solution: Sana is second and Omar is fifth, so Priya (third) and Rahul (fourth) sit between them, which is two runners. Counting only one or three misreads the gap, and zero ignores the block entirely.

Instructions (BS3-DILR-2): study the shared data below and answer its 3 question(s).

Four teams, Falcons, Hawks, Owls and Ravens, took part in a quiz. Their final scores were the four distinct values 30, 40, 50 and 60 points, in some order.

- The Hawks scored exactly 20 points more than the Owls. - The Ravens outscored the Falcons. - The Falcons did not finish with the lowest score.

Q112. [BS3-DILR-2 · mcsc · diff 0.5]

Which team won the quiz with the highest score?

- A) Hawks
- B) Owls
- C) Ravens ✓
- D) Falcons

Answer: C) Ravens

Solution: The Owls are not lowest only if Falcons take 30, but Falcons are not last, so the Owls score 30 and the Hawks score 50. That leaves 40 and 60 for Falcons and Ravens, and since Ravens outscore Falcons, Ravens take 60 and win. Hawks at 50 and Falcons at 40 are runners-up, and Owls are last.

Q113. [BS3-DILR-2 · mcsc · diff 0.5]

How many points did the Falcons score?

- A) 40 ✓
- B) 30
- C) 60
- D) 50

Answer: A) 40

Solution: With Owls on 30 and Hawks on 50, the remaining 40 and 60 go to Falcons and Ravens. Ravens outscore Falcons, so Falcons take 40. Reading 30 mistakes them for last, while 50 and 60 belong to Hawks and Ravens.

Q114. [BS3-DILR-2 · mcsc · diff 0.5]

What is the difference between the highest and the lowest team scores?

- A) 30 ✓
- B) 10
- C) 20
- D) 60

Answer: A) 30

Solution: The top score is the Ravens on 60 and the bottom is the Owls on 30, a gap of 30. The 20 trap echoes the Hawks-Owls margin, 60 mistakes the top score for the gap, and 10 halves the difference.

Instructions (BS3-DILR-3): study the shared data below and answer its 4 question(s).

A regional juice company sells three flavours, Mango, Guava and Litchi, at four riverside kiosks named P, Q, R and S. Table 1 gives the number of bottles sold and Table 2 gives the selling price of each bottle. All bottle counts are in thousands of bottles; all prices are in rupees per bottle; revenue = bottles sold x price per bottle, so each revenue figure comes out in thousands of rupees.

Table 1: Bottles sold (in thousands)

Flavour	P	Q	R	S
Mango	12	8	10	6
Guava	5	9	7	4
Litchi	8	6	5	9

Table 2: Price per bottle (in rupees)

Flavour	P	Q	R	S
Mango	25	25	30	30
Guava	20	20	20	20
Litchi	15	15	10	10

Q115. [BS3-DILR-3 · mcsc · diff 0.5]

How many bottles in total (in thousands) were sold at kiosk Q across all three flavours?

- A) 22 thousand
- B) 23 thousand ✓
- C) 21 thousand
- D) 24 thousand

Answer: B) 23 thousand

Solution: Kiosk Q is the second data column: Mango 8, Guava 9 and Litchi 6 bottles, all in thousands. Adding the column gives $8 + 9 + 6 = 23$ thousand bottles. The 21 trap drops the 8, the 22 trap misreads Guava as 8, and the 24 trap adds an extra unit.

Q116. [BS3-DILR-3 · mcsc · diff 0.5]

What was the total revenue (in thousands of rupees) generated by kiosk P?

- A) 540 thousand
- B) 470 thousand
- C) 520 thousand ✓
- D) 500 thousand

Answer: C) 520 thousand

Solution: At kiosk P: Mango $12 \times 25 = 300$, Guava $5 \times 20 = 100$ and Litchi $8 \times 15 = 120$, all in thousands of rupees. The total is $300 + 100 + 120 = 520$ thousand. The 500 and 540 traps mis-price the Litchi bottles and 470 is the revenue of kiosk Q instead of P.

Q117. [BS3-DILR-3 · mcsc · diff 0.5]

What was the combined revenue of the Mango flavour (in thousands of rupees) across all four kiosks?

- A) 880 thousand
- B) 1000 thousand
- C) 900 thousand
- D) 980 thousand ✓

Answer: D) 980 thousand

Solution: Mango revenue by kiosk: $12 \times 25 = 300$, $8 \times 25 = 200$, $10 \times 30 = 300$ and $6 \times 30 = 180$, all in thousands of rupees. The row total is $300 + 200 + 300 + 180 = 980$ thousand. The 900 and 880 traps apply a flat price of 25 or 30 to every kiosk, and 1000 rounds carelessly.

Q118. [BS3-DILR-3 · mcsc · diff 0.5]

In how many of the twelve flavour-kiosk cells did the revenue exceed 150 thousand rupees?

- A) 3 cells
- B) 4 cells
- C) 5 cells ✓
- D) 6 cells

Answer: C) 5 cells

Solution: Cell revenues in thousands are Mango 300, 200, 300, 180; Guava 100, 180, 140, 80; Litchi 120, 90, 50, 90. Those above 150 are the four Mango cells plus Guava at kiosk Q (180), giving 5 cells. The 4 trap forgets the Guava cell, while 6 and 3 miscount the Mango row.

Instructions (BS3-DILR-4): study the shared data below and answer its 3 question(s).

Four employees, Arjun, Bela, Carlos and Diana, each attended a one-day training session. The sessions were held on four different days: Monday, Tuesday, Wednesday and Thursday, one employee per day.

- Arjun trained on an earlier day than Bela. - Carlos trained exactly two days after Diana. - Diana did not train on Monday.

Q119. [BS3-DILR-4 · mcsc · diff 0.5]

On which day did Arjun train?

- A) Thursday
- B) Monday ✓
- C) Tuesday
- D) Wednesday

Answer: B) Monday

Solution: Carlos is two days after Diana, and Diana avoids Monday, so Diana is Tuesday and Carlos is Thursday. Monday and Wednesday remain for Arjun and Bela, and since Arjun is earlier, Arjun takes Monday. Tuesday and Thursday are taken by Diana and Carlos, and Wednesday belongs to Bela.

Q120. [BS3-DILR-4 · mcsc · diff 0.5]

Who trained on Wednesday?

- A) Diana
- B) Carlos
- C) Arjun
- D) Bela ✓

Answer: D) Bela

Solution: Diana is Tuesday and Carlos is Thursday, leaving Monday and Wednesday for Arjun and Bela. Arjun is earlier, so Arjun is Monday and Bela is Wednesday. Diana and Carlos are fixed on other days, and Arjun sits on Monday.

Q121. [BS3-DILR-4 · mcsc · diff 1]

Who trained on the day immediately before Carlos?

- A) Arjun
- B) No one, Carlos trained on the first day
- C) Bela ✓
- D) Diana

Answer: C) Bela

Solution: Carlos trains on Thursday, so the day immediately before is Wednesday, which is Bela. Diana trains two days earlier on Tuesday rather than the day before, Arjun is on Monday, and Carlos is certainly not on the first day.

Instructions (BS3-DILR-5): study the shared data below and answer its 5 question(s).

The grouped bar chart below shows the number of units sold by three solar lamp models, Solaris, Lumen and Vireo, in the years 2022, 2023 and 2024. All figures are in thousands of units, so a bar of height 40 means 40 thousand units were sold.

Q122. [BS3-DILR-5 · mcsc · diff 0.5]

How many units in total (in thousands) were sold across all three models in 2023?

- A) 115 thousand
- B) 130 thousand
- C) 110 thousand
- D) 120 thousand ✓

Answer: D) 120 thousand

Solution: The 2023 bars read 50 for Solaris, 40 for Lumen and 30 for Vireo, all in thousands of units. Their sum is $50 + 40 + 30 = 120$ thousand. The 110 trap drops a Vireo unit, 130 borrows a 2024 bar, and 115 comes from misreading Solaris as 45.

Q123. [BS3-DILR-5 · mcsc · diff 0.5]

By what percentage did the units sold of Solaris grow from 2022 to 2024?

- A) 50 percent ✓
- B) 33 percent
- C) 20 percent
- D) 40 percent

Answer: A) 50 percent

Solution: Solaris rose from 40 thousand units in 2022 to 60 thousand in 2024, an increase of 20 thousand. As a percentage of the 2022 base, $20 / 40 = 50$ percent. The 33 trap wrongly uses 60 as the base, 20 is the raw increase read as a percentage, and 40 mixes up the models.

Q124. [BS3-DILR-5 · mcsc · diff 0.5]

In which year were the total units sold across all three models the highest?

- A) Equal in 2023 and 2024
- B) 2022
- C) 2024 ✓
- D) 2023

Answer: C) 2024

Solution: Year totals in thousands are $40 + 30 + 20 = 90$ for 2022, $50 + 40 + 30 = 120$ for 2023 and $60 + 50 + 40 = 150$ for 2024. The largest total, 150 thousand, belongs to 2024. Because 150 clearly beats 120, the years are not tied and neither earlier year leads.

Q125. [BS3-DILR-5 · mcsc · diff 0.5]

What is the ratio of Lumen units sold in 2024 to Vireo units sold in 2022?

- A) 5 : 2 ✓
- B) 5 : 3
- C) 3 : 1
- D) 2 : 1

Answer: A) 5 : 2

Solution: Lumen sold 50 thousand units in 2024 and Vireo sold 20 thousand units in 2022. The ratio $50 : 20$ divides by 10 to give $5 : 2$. The $2 : 1$ and $5 : 3$ traps come from misreading one of the bars, and $3 : 1$ wrongly pairs Lumen with a 2023 Vireo value.

Q126. [BS3-DILR-5 · mcsc · diff 0.5]

What was the average number of units (in thousands) sold per model in 2024?

- A) 45 thousand
- B) 48 thousand
- C) 50 thousand ✓
- D) 55 thousand

Answer: C) 50 thousand

Solution: The 2024 bars are 60 for Solaris, 50 for Lumen and 40 for Vireo, all in thousands of units, summing to 150 thousand. Dividing by the three models gives $150 / 3 = 50$ thousand. The 45 trap averages the 2023 figures, and 55 and 48 come from arithmetic slips on the total.

Section: QA (40 min)

Q127. [BS1-QA-MCQ-1 · mcsc · diff 0]

A shopkeeper buys an article for 80 euros and sells it for 100 euros. What is the profit percentage?

- A) 20%
- B) 80%
- C) 125%
- D) 25% ✓

Answer: D) 25%

Q128. [BS1-QA-MCQ-2 · mcsc · diff 0]

The ratio of boys to girls in a class is 3 : 5. If there are 20 girls, how many boys are there?

- A) 33 boys
- B) 15 boys
- C) 8 boys
- D) 12 boys ✓

Answer: D) 12 boys

Q129. [BS1-QA-MCQ-3 · mcsc · diff 0]

What is the average of the five numbers 4, 8, 10, 12 and 16?

- A) 10 ✓
- B) 50
- C) 12
- D) 8

Answer: A) 10

Q130. [BS1-QA-MCQ-4 · mcsc · diff 0]

A circle has radius 7 cm. What is its area? (use $\pi = 22/7$)

- A) 44 cm^2
- B) 154 cm^2 ✓
- C) 616 cm^2
- D) 49 cm^2

Answer: B) 154 cm^2

Q131. [BS1-QA-MCQ-5 · mcsc · diff 0]

Solve for x: $3x + 7 = 22$.

- A) $29/3$
- B) 9
- C) 5 ✓
- D) 15

Answer: C) 5

Q132. [BS1-QA-MCQ-6 · mcsc · diff 0]

A car travels 150 km in 3 hours. What is its average speed?

- A) 50 km/h ✓
- B) 450 km/h
- C) 75 km/h
- D) 2 km/h

Answer: A) 50 km/h

Q133. [BS1-QA-MCQ-7 · mcsc · diff 0.5]

An article marked at 500 euros is sold after a 20% discount, and the seller still makes a 25% profit on cost. What was the cost price?

- A) 400 euros
- B) 333 euros
- C) 320 euros ✓
- D) 300 euros

Answer: C) 320 euros

Q134. [BS1-QA-MCQ-8 · mcsc · diff 0.5]

A sum of 8000 euros is invested at 10% per annum compound interest. What is the amount after 2 years?

- A) 1680 euros
- B) 9700 euros
- C) 9600 euros
- D) 9680 euros ✓

Answer: D) 9680 euros

Q135. [BS1-QA-MCQ-9 · mcsc · diff 0.5]

Pipe A fills a tank in 12 hours and pipe B fills it in 6 hours. If both pipes are opened together, how long do they take to fill the tank?

- A) 4 hours ✓
- B) 9 hours
- C) 18 hours
- D) 3 hours

Answer: A) 4 hours

Q136. [BS1-QA-MCQ-10 · mcsc · diff 0.5]

When a positive integer n is divided by 7 the remainder is 4. What is the remainder when $3n$ is divided by 7?

- A) 4
- B) 12
- C) 2
- D) 5 ✓

Answer: D) 5

Q137. [BS1-QA-MCQ-11 · mcsc · diff 0.5]

The HCF of two numbers is 6 and their LCM is 180. If one number is 36, what is the other number?

- A) 30 ✓
- B) 180
- C) 1080
- D) 5

Answer: A) 30

Q138. [BS1-QA-MCQ-12 · mcsc · diff 0.5]

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both are red?

A) $\frac{2}{15}$ ✓

B) $\frac{4}{25}$

C) $\frac{1}{5}$

D) $\frac{2}{9}$

Answer: A) $\frac{2}{15}$

Q139. [BS1-QA-MCQ-13 · mcsc · diff 0.5]

How many distinct arrangements can be made using all the letters of the word LEVEL?

A) 120

B) 30 ✓

C) 60

D) 20

Answer: B) 30

Q140. [BS1-QA-MCQ-14 · mcsc · diff 0.5]

In a group of 50 people, 30 like tea and 25 like coffee. If 10 like neither, how many like both tea and coffee?

A) 5

B) 15 ✓

C) 55

D) 40

Answer: B) 15

Q141. [BS1-QA-MCQ-15 · mcsc · diff 0.5]

The third term of an arithmetic progression is 11 and the seventh term is 27. What is the first term?

A) 5

B) 4

C) 7

D) 3 ✓

Answer: D) 3

Q142. [BS1-QA-MCQ-16 · mcsc · diff 1]

Two trains start at the same time from stations A and B, 330 km apart, moving towards each other at 60 km/h and 50 km/h. How far from A do they meet?

A) 180 km ✓

B) 150 km

C) 165 km

D) 110 km

Answer: A) 180 km

Q143. [BS1-QA-MCQ-17 · mcsc · diff 1]

Vessel A holds 40 litres of a mixture with milk and water in the ratio 3 : 1, and vessel B holds 20 litres in the ratio 1 :

1. If both are poured into one vessel, what is the ratio of milk to water in the new mixture?

A) 2 : 1 ✓

B) 4 : 3

C) 5 : 3

D) 3 : 2

Answer: A) 2 : 1

Q144. [BS1-QA-MCQ-18 · mcsc · diff 1]

A rectangular garden is 40 m long and 30 m wide. A path of uniform width 2 m runs all around it on the inside. What is the area of the path?

- A) 280 m²
- B) 144 m²
- C) 264 m² ✓
- D) 536 m²

Answer: C) 264 m²

Q145. [BS1-QA-TITA-1 · tita · diff 0]

Evaluate the expression $(2^5 \times 2^3) \div 2^4$. Type your answer as an integer.

Answer (TITA): 16

Q146. [BS1-QA-TITA-2 · tita · diff 0]

An item is priced at 80 euros. Its price is first increased by 25%, then the new price is decreased by 20%. What is the final price, in euros? Type your answer as an integer.

Answer (TITA): 80

Q147. [BS1-QA-TITA-3 · tita · diff 0.5]

Using the digits 1, 2, 3, 4 and 5 without repetition, how many three-digit numbers divisible by 5 can be formed? Type your answer as an integer.

Answer (TITA): 12

Q148. [BS1-QA-TITA-4 · tita · diff 0.5]

In a class, 30 students study French and 25 study German. Exactly 10 students study both languages. How many students study exactly one of the two languages? Type your answer as an integer.

Answer (TITA): 35

Q149. [BS1-QA-TITA-5 · tita · diff 0.5]

In an arithmetic progression the first term is 7 and the common difference is 5. What is the value of the 12th term? Type your answer as an integer.

Answer (TITA): 62

Q150. [BS1-QA-TITA-6 · tita · diff 1]

A vessel contains 64 litres of pure wine. 16 litres are drawn out and replaced with water; this operation (drawing 16 litres of the mixture and replacing it with water) is performed two more times, for a total of three times. How many litres of pure wine remain in the vessel? Type your answer as an integer.

Answer (TITA): 27

Q151. [BS2-QA-MCQ-1 · mcsc · diff 0]

A school has 640 students, and 35% of them travel to school by bus. How many students travel by bus?

- A) 192
- B) 416
- C) 448
- D) 224 ✓

Answer: D) 224

Q152. [BS2-QA-MCQ-2 · mcsc · diff 0]

A trader buys a chair for 250 euros and sells it at a profit of 15% on the cost price. What is the selling price, in euros?

- A) 265.00
- B) 275.00
- C) 212.50
- D) 287.50 ✓

Answer: D) 287.50

Q153. [BS2-QA-MCQ-3 · mcsc · diff 0]

A sum of 3300 euros is divided between Anna and Bruno in the ratio 4 : 7. How many euros does Bruno receive?

- A) 2100 ✓
- B) 1650
- C) 2200
- D) 1200

Answer: A) 2100

Q154. [BS2-QA-MCQ-4 · mcsc · diff 0]

Six readings taken during an experiment add up to 261. What is the average of the six readings?

- A) 37.3
- B) 52.2
- C) 44.5
- D) 43.5 ✓

Answer: D) 43.5

Q155. [BS2-QA-MCQ-5 · mcsc · diff 0]

A circular flower bed has a diameter of 14 metres. What is the length of the fence needed to enclose it exactly once? (take $\pi = 22/7$)

- A) 22 m
- B) 154 m
- C) 88 m
- D) 44 m ✓

Answer: D) 44 m

Q156. [BS2-QA-MCQ-6 · mcsc · diff 0]

Two bells ring at intervals of 15 seconds and 18 seconds. If they ring together once, after how many seconds will they next ring together?

- A) 45
- B) 270
- C) 180
- D) 90 ✓

Answer: D) 90

Q157. [BS2-QA-MCQ-7 · mcsc · diff 0.5]

A sum of 12000 euros is lent at 8% per annum simple interest. What total amount, in euros, is repaid at the end of 2 years and 6 months?

- A) 14400 ✓
- B) 13920
- C) 14880
- D) 2400

Answer: A) 14400

Q158. [BS2-QA-MCQ-8 · mcsc · diff 0.5]

A cyclist rides from a village to a market 30 km away at 15 km/h and returns along the same road at 10 km/h. What is the average speed for the whole trip?

- A) 12.5 km/h
- B) 25 km/h
- C) 12 km/h ✓
- D) 11 km/h

Answer: C) 12 km/h

Q159. [BS2-QA-MCQ-9 · mcsc · diff 0.5]

A water tank is $\frac{3}{5}$ full. After 24 litres are drawn out, the tank is $\frac{1}{3}$ full. What is the full capacity of the tank, in litres?

- A) 40
- B) 120
- C) 72
- D) 90 ✓

Answer: D) 90

Q160. [BS2-QA-MCQ-10 · mcsc · diff 0.5]

A sum of 12000 euros is lent for 2 years at 10% per annum. By how many euros does the compound interest, compounded annually, exceed the simple interest for the same period?

- A) 120 euros ✓
- B) 240 euros
- C) 252 euros
- D) 100 euros

Answer: A) 120 euros

Q161. [BS2-QA-MCQ-11 · mcsc · diff 0.5]

Tap A alone can fill an empty tank in 15 minutes, while an outlet pipe B alone can empty a full tank in 20 minutes. If both are opened together when the tank is empty, how long will the tank take to fill completely?

- A) 60 minutes ✓
- B) 35 minutes
- C) 45 minutes
- D) 30 minutes

Answer: A) 60 minutes

Q162. [BS2-QA-MCQ-12 · mcsc · diff 0.5]

A laboratory has 40 litres of a solution in which acid forms 15% of the volume. How many litres of pure water must be added so that acid forms exactly 10% of the resulting solution?

- A) 5 litres
- B) 10 litres
- C) 15 litres
- D) 20 litres ✓

Answer: D) 20 litres

Q163. [BS2-QA-MCQ-13 · mcsc · diff 0.5]

A box holds 5 red marbles and 3 green marbles. One marble is drawn at random, its colour is noted, and it is put back before a second marble is drawn. What is the probability that the two marbles drawn have the same colour?

- A) $\frac{17}{32}$ ✓
- B) $\frac{1}{2}$
- C) $\frac{13}{28}$
- D) $\frac{15}{64}$

Answer: A) $\frac{17}{32}$

Q164. [BS2-QA-MCQ-14 · mcsc · diff 0.5]

A committee of 3 members is to be chosen from 5 men and 4 women. In how many ways can this be done if the committee must include at least one woman?

- A) 64
- B) 74 ✓
- C) 84
- D) 40

Answer: B) 74

Q165. [BS2-QA-MCQ-15 · mcsc · diff 0.5]

A closed solid cylinder has a base radius of 7 cm and a height of 10 cm. Taking $\pi = 22/7$, what is its total surface area?

- A) 594 square cm
- B) 440 square cm
- C) 902 square cm
- D) 748 square cm ✓

Answer: D) 748 square cm

Q166. [BS2-QA-MCQ-16 · mcsc · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by each of 6, 7 and 8, and is also exactly divisible by 11?

- A) 1683 ✓
- B) 1680
- C) 2523
- D) 171

Answer: A) 1683

Q167. [BS2-QA-MCQ-17 · mcsc · diff 1]

Working in pairs, A and B finish a job in 12 days, B and C finish the same job in 15 days, and A and C finish it in 20 days. Working all three together, in how many days will they finish the job?

- A) 12 days
- B) 5 days
- C) 10 days ✓
- D) 16 days

Answer: C) 10 days

Q168. [BS2-QA-MCQ-18 · mcsc · diff 1]

In an arithmetic progression the sum of the first 15 terms is 600 and the 5th term is 22. What is the 20th term of this progression?

- A) 112 ✓
- B) 118
- C) 100
- D) 106

Answer: A) 112

Q169. [BS2-QA-TITA-1 · tita · diff 0]

In a geometric progression the first term is 5 and the common ratio is 3. What is the fourth term? Type your answer as an integer.

Answer (TITA): 135

Q170. [BS2-QA-TITA-2 · tita · diff 0]

What is the sum of all even numbers from 2 to 40, both included? Type your answer as an integer.

Answer (TITA): 420

Q171. [BS2-QA-TITA-3 · tita · diff 0.5]

The population of a town is 12500. During the first year it rises by 20%, and during the second year it falls by 10%. What is the population at the end of the second year? Type your answer as an integer.

Answer (TITA): 13500

Q172. [BS2-QA-TITA-4 · tita · diff 0.5]

How many four-digit numbers can be formed using the digits 0, 1, 2, 3 and 4 if no digit is repeated in a number? Type your answer as an integer.

Answer (TITA): 96

Q173. [BS2-QA-TITA-5 · tita · diff 0.5]

A train 180 m long crosses a platform 240 m long in 21 seconds. What is the speed of the train in km/h? Type your answer as an integer.

Answer (TITA): 72

Q174. [BS2-QA-TITA-6 · tita · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by 5, a remainder of 4 when divided by 6, and a remainder of 5 when divided by 7? Type your answer as an integer.

Answer (TITA): 208

Q175. [BS3-QA-MCQ-1 · mcsc · diff 0]

In a class of 250 students, 36% opted for the science club. How many students opted for the science club?

- A) 90 ✓
- B) 214
- C) 160
- D) 36

Answer: A) 90

Solution: 36% of 250 equals $0.36 \times 250 = 90$ students, so 90 is correct. The value 36 wrongly reads the percentage as the count. The value 160 counts the remaining 64% who did not opt in, and 214 subtracts 36 from 250 by treating the percent as an absolute figure.

Q176. [BS3-QA-MCQ-2 · mcsc · diff 0]

A shopkeeper buys a bag for 400 rupees and sells it for 460 rupees. What is the profit percentage?

- A) 13%
- B) 15% ✓
- C) 12%
- D) 60%

Answer: B) 15%

Solution: Profit is $460 - 400 = 60$, and profit percent is $60/400 \times 100 = 15\%$, so 15% is correct. The 13% figure wrongly uses the selling price 460 as the base. The 60% reads the absolute profit of 60 rupees as a percentage, and 12% divides 60 by 500.

Q177. [BS3-QA-MCQ-3 · mcsc · diff 0]

Two numbers are in the ratio 3 : 5, and their sum is 96. What is the larger number?

- A) 48
- B) 36
- C) 60 ✓
- D) 58

Answer: C) 60

Solution: The total of parts is $3 + 5 = 8$, so each part is $96/8 = 12$ and the larger number is $5 \times 12 = 60$. The 36 value gives the smaller number 3×12 . The 48 splits the sum evenly in half, and 58 comes from an incorrect near-half split.

Q178. [BS3-QA-MCQ-4 · mcsc · diff 0]

The average of five numbers is 24. What is their total sum?

- A) 24
- B) 120 ✓
- C) 100
- D) 29

Answer: B) 120

Solution: The sum equals the average times the count, so $24 \times 5 = 120$, making 120 correct. The 29 value wrongly adds the count 5 to the average. The 24 mistakes the average itself for the sum, and 100 uses an incorrect count of about 4.

Q179. [BS3-QA-MCQ-5 · mcsc · diff 0]

A sum of 2000 rupees is invested at 6% simple interest per year. What is the interest earned in one year?

- A) 120 ✓
- B) 2120
- C) 60
- D) 112

Answer: A) 120

Solution: Simple interest for one year is $2000 \times 6/100 = 120$ rupees, so 120 is correct. The 2120 value adds the principal to the interest, giving the maturity amount. The 60 halves the correct interest, and 112 uses a rate near 5.6%.

Q180. [BS3-QA-MCQ-6 · mcsc · diff 0]

A rectangular garden is 14 metres long and 9 metres wide. What is its perimeter?

- A) 52
- B) 126
- C) 23
- D) 46 ✓

Answer: D) 46

Solution: Perimeter of a rectangle is $2 \times (\text{length} + \text{width}) = 2 \times (14 + 9) = 46$ metres, so 46 is correct. The 126 value computes the area 14×9 instead. The 23 gives just length plus width without doubling, and 52 uses an incorrect width of 12.

Q181. [BS3-QA-MCQ-7 · mcsc · diff 0.5]

A car travels 165 kilometres in 3 hours, then 120 kilometres in 2 hours. What is its average speed for the whole journey?

- A) 55
- B) 57 ✓
- C) 60
- D) 56

Answer: B) 57

Solution: Total distance is $165 + 120 = 285$ km over $3 + 2 = 5$ hours, so average speed is $285/5 = 57$ km/h, making 57 correct. The 60 value averages the two segment speeds of 55 and 60 wrongly weighted. The 55 uses only the first segment speed, and 56 averages the two speeds arithmetically.

Q182. [BS3-QA-MCQ-8 · mcsc · diff 0.5]

The smallest number that is exactly divisible by 8, 12, and 18 is:

- A) 144
- B) 36
- C) 24
- D) 72 ✓

Answer: D) 72

Solution: The LCM of 8, 12, and 18 is found from prime factors 2^3 , $2^2 \times 3$, and 2×3^2 , giving $2^3 \times 3^2 = 72$, so 72 is correct. The 36 value is only the LCM of 12 and 18 and misses the factor of 8. The 144 doubles the answer unnecessarily, and 24 is only the LCM of 8 and 12 while ignoring 18.

Q183. [BS3-QA-MCQ-9 · mcsc · diff 0.5]

A shirt is marked at 800 rupees. After a discount of 15%, the shopkeeper still makes a profit of 20% on the cost price. What was the cost price?

- A) 567 ✓
- B) 600
- C) 680
- D) 544

Answer: A) 567

Solution: The selling price after 15% discount is $800 \times 0.85 = 680$, and this equals 1.2 times the cost, so cost is $680/1.2 = 567$ rupees approximately, making 567 correct. The 544 wrongly takes 20% off 680. The 680 mistakes the selling price for the cost, and 600 uses a $800/1.333$ division.

Q184. [BS3-QA-MCQ-10 · mcsc · diff 0.5]

A sum of Rs 8000 is invested at 10% per annum compound interest, compounded annually. What is the compound interest earned at the end of 2 years?

- A) Rs 1600
- B) Rs 968
- C) Rs 1728
- D) Rs 1680 ✓

Answer: D) Rs 1680

Solution: Amount after 2 years is $8000 \times 1.1 \times 1.1 = 9680$, so the compound interest is $9680 - 8000 = 1680$. The Rs 1600 trap uses simple interest (800 per year). Rs 1728 wrongly compounds for 3 years, and Rs 968 takes only the second-year interest.

Q185. [BS3-QA-MCQ-11 · mcsc · diff 0.5]

Pipe A can fill a tank in 12 hours and pipe B can fill the same tank in 18 hours. If both pipes are opened together, how long will they take to fill the tank?

- A) 7.5 hours
- B) 15 hours
- C) 6 hours
- D) 7.2 hours ✓

Answer: D) 7.2 hours

Solution: Combined rate is $1/12 + 1/18 = 3/36 + 2/36 = 5/36$ tank per hour, so the time is $36/5 = 7.2$ hours. The 15-hour value wrongly adds the two times, 6 hours halves the smaller time, and 7.5 comes from averaging 12 and 18 incorrectly.

Q186. [BS3-QA-MCQ-12 · mcsc · diff 0.5]

A 40-litre mixture contains milk and water in the ratio 3:1. How many litres of water must be added so that the ratio of milk to water becomes 3:2?

- A) 5 litres
- B) 10 litres ✓
- C) 20 litres
- D) 8 litres

Answer: B) 10 litres

Solution: The mixture has 30 litres of milk and 10 litres of water. Milk stays fixed, so $30/(10 + x) = 3/2$ gives $10 + x = 20$, hence $x = 10$ litres. The 5-litre answer targets a 3:1.5 split, 8 litres misreads the target ratio, and 20 litres doubles the water to reach 1:1.

Q187. [BS3-QA-MCQ-13 · mcsc · diff 0.5]

Two fair six-sided dice are rolled together. What is the probability that the sum of the two numbers shown is exactly 8?

- A) 1/9
- B) 5/36 ✓
- C) 1/6
- D) 7/36

Answer: B) 5/36

Solution: The favourable outcomes are (2,6), (3,5), (4,4), (5,3) and (6,2), which is 5 of the 36 equally likely pairs, so the probability is 5/36. The 7/36 trap over-counts the pairs, 1/6 assumes 6 outcomes, and 1/9 counts only 4.

Q188. [BS3-QA-MCQ-14 · mcsc · diff 0.5]

In how many distinct ways can all the letters of the word BANANA be arranged?

- A) 720
- B) 120
- C) 360
- D) 60 ✓

Answer: D) 60

Solution: BANANA has 6 letters with three A's and two N's repeated, so the count is $6!/(3! \times 2!) = 720/12 = 60$. The 720 value forgets all repeats, 120 divides by 3! only, and 360 divides by 2! only.

Q189. [BS3-QA-MCQ-15 · mcsc · diff 0.5]

A solid sphere has a radius of 3 cm. What is its volume?

- A) $36\pi \text{ cm}^3$ ✓
- B) $27\pi \text{ cm}^3$
- C) $12\pi \text{ cm}^3$
- D) $9\pi \text{ cm}^3$

Answer: A) $36\pi \text{ cm}^3$

Solution: Volume of a sphere is $(4/3)\pi r^3 = (4/3)\pi(27) = 36\pi$ cubic cm. The 27π value drops the $4/3$ factor, 12π uses r^2 instead of r^3 , and 9π confuses volume with a surface-related figure.

Q190. [BS3-QA-MCQ-16 · mcsc · diff 1]

A certain sum invested at compound interest, compounded annually, amounts to Rs 4840 in 2 years and to Rs 5324 in 3 years. What was the original sum invested?

- A) Rs 3993
- B) Rs 4400
- C) Rs 4000 ✓
- D) Rs 4200

Answer: C) Rs 4000

Solution: The third-year interest is $5324 - 4840 = 484$ on 4840, giving a rate of $484/4840 = 10\%$. Then the principal satisfies $P \times 1.1 \times 1.1 = 4840$, so $P = 4840/1.21 = 4000$. The Rs 4400 value is the amount after 1 year, Rs 4200 subtracts interest linearly, and Rs 3993 mis-applies a 3-year discount.

Q191. [BS3-QA-MCQ-17 · mcsc · diff 1]

In an arithmetic progression the third term is 12 and the seventh term is 24. What is the sum of the first 10 terms of this progression?

- A) 195 ✓
- B) 165
- C) 210
- D) 180

Answer: A) 195

Solution: From $a + 2d = 12$ and $a + 6d = 24$, subtracting gives $4d = 12$ so $d = 3$ and $a = 6$. The sum of 10 terms is $(10/2)(2a + 9d) = 5(12 + 27) = 5 \times 39 = 195$. The 210 value uses $d = 3$ but $a = 9$, 180 forgets the $9d$ part, and 165 uses only $8d$ in the bracket.

Q192. [BS3-QA-MCQ-18 · mcsc · diff 1]

In a group of 120 students, 70 like tea, 80 like coffee, and 20 like neither tea nor coffee. How many students like both tea and coffee?

- A) 40
- B) 30
- C) 50 ✓
- D) 60

Answer: C) 50

Solution: The number liking at least one drink is $120 - 20 = 100$. By inclusion-exclusion, both = $70 + 80 - 100 = 50$. The 30 trap forgets to remove the 20 who like neither, 40 subtracts 20 twice, and 60 mis-adds the totals.

Q193. [BS3-QA-TITA-1 · tita · diff 0]

Find the value of the expression $(2^5 \times 3^2)$ divided by $(2^3 \times 3)$.

Answer (TITA): 12

Solution: Subtract exponents of like bases: $2^{(5-3)} \times 3^{(2-1)}$ equals $2^2 \times 3^1$, which is 4×3 equals 12. A reader who forgets to reduce the powers of 3 and writes 4 lands short, while cancelling only the twos gives 4×9 equals 36.

Q194. [BS3-QA-TITA-2 · tita · diff 0]

A train travels at a constant speed of 72 kilometres per hour. Express this speed in metres per second.

Answer (TITA): 20

Solution: To convert kilometres per hour to metres per second, multiply by 5 and divide by 18: $72 \times 5 / 18$ equals $360 / 18$ equals 20 metres per second. Multiplying by $18/5$ instead of $5/18$ gives the wrong figure of about 259, a common inversion slip.

Q195. [BS3-QA-TITA-3 · tita · diff 0.5]

In an election between two candidates, the winner secured 60 percent of the total valid votes and won by a margin of 4800 votes. Find the total number of valid votes polled.

Answer (TITA): 24000

Solution: The winner got 60 percent and the loser 40 percent, so the margin is 20 percent of the total. If 20 percent equals 4800, the total is $4800 \times 100 / 20$ equals 24000. Treating 4800 as the winner's 60 percent share instead of the 20 percent margin gives a wrong total of 8000.

Q196. [BS3-QA-TITA-4 · tita · diff 0.5]

How many positive integer factors does 180 have?

Answer (TITA): 18

Solution: 180 factors as $2^2 \times 3^2 \times 5^1$, so the number of positive factors is $(2+1)(2+1)(1+1) = 3 \times 3 \times 2 = 18$. Counting only 1, 180 and the primes, or forgetting a prime, undercounts; squaring an exponent overcounts.

Q197. [BS3-QA-TITA-5 · tita · diff 0.5]

In a group of students, 60 like tea, 50 like coffee, and 20 like both tea and coffee. How many students like exactly one of the two drinks?

Answer (TITA): 70

Solution: Students liking only tea number 60 minus 20 equals 40, and those liking only coffee number 50 minus 20 equals 30, so exactly one is 40 plus 30 equals 70. Adding 60 and 50 to get 110 counts the 20 who like both twice and answers the wrong question.

Q198. [BS3-QA-TITA-6 · tita · diff 1]

Find the remainder when 7 raised to the power 103 is divided by 100.

Answer (TITA): 43

Solution: The last two digits of powers of 7 cycle with period 4 as 07, 49, 43, 01 and then repeat. Since 103 leaves remainder 3 when divided by 4, the value matches the third term of the cycle, which is 43. Taking 103 modulo 4 as 0 instead of 3 would wrongly point to the ending 01.